



US012401655B1

(12) **United States Patent**
Peter, III et al.

(10) **Patent No.: US 12,401,655 B1**
(45) **Date of Patent: Aug. 26, 2025**

(54) **SYSTEMS AND METHODS TO MANAGE
ACCESS TO ASSETS OF A COMPUTER
ENVIRONMENT BASED ON USER AND
ASSET GROUPING**

(71) Applicant: **Asana, Inc.**, San Francisco, CA (US)

(72) Inventors: **Phillips Smith Peter, III**, Long Island
City, NY (US); **Caleb David Wang**,
San Francisco, CA (US); **Joshua Ryan
Smith**, Albany, NY (US); **Megan
Maureen Daly**, San Francisco, CA
(US); **Zachary Isaac Millman**, San
Francisco, CA (US)

(73) Assignee: **Asana, Inc.**, San Francisco, CA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 170 days.

(21) Appl. No.: **18/305,763**

(22) Filed: **Apr. 24, 2023**

(51) **Int. Cl.**
H04L 9/40 (2022.01)

(52) **U.S. Cl.**
CPC **H04L 63/104** (2013.01); **H04L 63/105**
(2013.01)

(58) **Field of Classification Search**
CPC H04L 63/104; H04L 63/105
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,233,687	A	8/1993	Henderson, Jr.
5,524,077	A	6/1996	Faaland
5,530,861	A	6/1996	Diamant
5,608,898	A	3/1997	Turpin

5,611,076	A	3/1997	Durflinger
5,623,404	A	4/1997	Collins
5,721,770	A	2/1998	Kohler
5,983,277	A	11/1999	Heile
5,987,453	A	11/1999	Krishna
6,024,093	A	2/2000	Cron
6,167,399	A	12/2000	Hoang
6,256,651	B1	7/2001	Tuli
6,292,830	B1	9/2001	Taylor
6,332,147	B1	12/2001	Moran
6,385,639	B1	5/2002	Togawa
6,442,748	B1	8/2002	Bowman-Amuah

(Continued)

FOREIGN PATENT DOCUMENTS

CN	101305350	A	11/2008
CN	101563671	A	10/2009

(Continued)

OTHER PUBLICATIONS

“U.S. Appl. No. 14/584,750, Examiner Interview Summary mailed
Feb. 25, 2016”, 3 pgs.

(Continued)

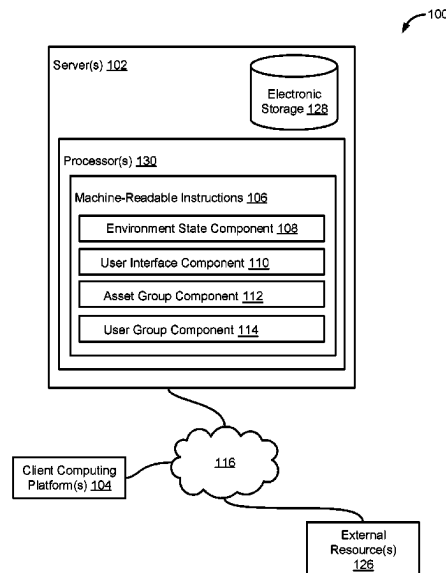
Primary Examiner — Michael Won

(74) *Attorney, Agent, or Firm* — Esplin & Associates, PC

(57) **ABSTRACT**

Systems and methods to manage access to assets of a
computer environment based on user and asset grouping are
described herein. Exemplary implementations may perform
one or more of: manage asset groups associated with one or
more assets of a computer environment; manage user groups
associated with individual asset groups; assign users to user
groups to cause access characteristics of associated assets to
be granted to users; enable and/or disable individual access
characteristics based on assigned ones of the user groups;
and/or other operations.

14 Claims, 4 Drawing Sheets



US 12,401,655 B1

Page 2

(56)	References Cited				10,594,788	B2	3/2020	Larabie-Belanger	
U.S. PATENT DOCUMENTS				10,606,859	B2	3/2020	Smith		
				10,606,901	B1	3/2020	Nair		
				10,613,735	B1	4/2020	Karpe		
6,621,505	B1	9/2003	Beauchamp	10,616,151	B1	4/2020	Cameron		
6,629,081	B1	9/2003	Cornelius	10,623,359	B1	4/2020	Rosenstein		
6,769,013	B2	7/2004	Frees	10,671,692	B2	6/2020	Koopman		
6,859,523	B1	2/2005	Jilk	10,684,870	B1	6/2020	Sabo		
7,020,697	B1	3/2006	Goodman	10,706,484	B1	7/2020	Murnock		
7,039,596	B1	5/2006	Lu	10,785,046	B1	9/2020	Raghavan		
7,086,062	B1	8/2006	Faour	10,810,222	B2	10/2020	Koch		
7,143,108	B1	11/2006	George	10,846,105	B2	11/2020	Granot		
7,349,920	B1	3/2008	Feinberg	10,846,297	B2	11/2020	Smith		
7,386,797	B1	6/2008	Chatterjee	10,922,104	B2	2/2021	Sabo		
7,418,482	B1	8/2008	Lusher	10,956,845	B1	3/2021	Sabo		
7,428,723	B2	9/2008	Greene	10,970,299	B2	4/2021	Smith		
7,640,511	B1	12/2009	Keel	10,977,434	B2	4/2021	Pelz		
7,676,542	B2	3/2010	Moser	10,983,685	B2	4/2021	Karpe		
7,779,039	B2	8/2010	Weissman	11,082,281	B2	8/2021	Justin		
7,801,886	B1	9/2010	Gabriel	11,095,468	B1	8/2021	Pandey		
7,805,327	B1	9/2010	Schulz	11,108,828	B1 *	8/2021	Curtis	H04L 63/104	
RE41,848	E	10/2010	Daniell	11,113,667	B1	9/2021	Jiang		
7,917,855	B1	3/2011	Satish	11,138,021	B1	10/2021	Rosenstein		
7,996,744	B2	8/2011	Ojala	11,140,174	B2	10/2021	Patel		
7,996,774	B1	8/2011	Sidenur	11,204,683	B1	12/2021	Sabo		
8,214,747	B1	7/2012	Yankovich	11,212,242	B2	12/2021	Cameron		
8,314,809	B1	11/2012	Grabowski	11,263,228	B2	3/2022	Koch		
8,327,419	B1 *	12/2012	Korablev	11,288,081	B2	3/2022	Sabo		
				11,290,296	B2	3/2022	Raghavan		
8,499,300	B2	7/2013	Zimberg	11,327,645	B2	5/2022	Karpe		
8,522,240	B1	8/2013	Merwarth	11,341,444	B2	5/2022	Sabo		
8,527,287	B1	9/2013	Bhatia	11,341,445	B1	5/2022	Cheng		
8,554,832	B1	10/2013	Moskovitz	2002/0065798	A1	5/2002	Bostleman		
8,572,477	B1	10/2013	Moskovitz	2002/0082889	A1	6/2002	Oliver		
8,627,199	B1	1/2014	Handley	2002/0143594	A1	10/2002	Kroeger		
8,639,552	B1	1/2014	Chen	2003/0028595	A1	2/2003	Vogt		
8,700,537	B1	4/2014	Deshpande	2003/0028616	A1	2/2003	Aoki		
8,768,751	B2	7/2014	Jakowski	2003/0036934	A1	2/2003	Ouchi		
8,831,879	B2	9/2014	Stamm	2003/0041317	A1	2/2003	Sokolov		
8,843,832	B2	9/2014	Frields	2003/0046576	A1 *	3/2003	High, Jr.	H04L 63/20	
8,863,021	B1	10/2014	Bee					726/1	
9,009,096	B2	4/2015	Pinckney	2003/0097406	A1	5/2003	Stafford		
9,024,752	B2	5/2015	Tumayan	2003/0097410	A1	5/2003	Atkins		
9,143,839	B2	9/2015	Reisman	2003/0126001	A1	7/2003	Northcutt		
9,152,668	B1	10/2015	Moskovitz	2003/0156138	A1	8/2003	Vronay		
9,201,952	B1	12/2015	Chau	2003/0200223	A1	10/2003	Hack		
9,208,262	B2	12/2015	Bechtel	2003/0220975	A1 *	11/2003	Malik	H04L 63/105	
9,251,484	B2	2/2016	Cantor					709/205	
9,350,560	B2	5/2016	Hupfer	2003/0225598	A1	12/2003	Yu		
9,383,917	B2	7/2016	Mouton	2003/0233265	A1	12/2003	Lee		
9,405,532	B1	8/2016	Sullivan	2003/0233268	A1	12/2003	Taqbeem		
9,405,810	B2	8/2016	Smith	2003/0236979	A1 *	12/2003	Himmel	H04L 63/08	
9,454,623	B1	9/2016	Kaptsan					713/167	
9,514,424	B2	12/2016	Kleinbart	2004/0083448	A1	4/2004	Schulz		
9,600,136	B1	3/2017	Yang	2004/0093290	A1	5/2004	Doss		
9,674,361	B2	6/2017	Ristock	2004/0093351	A1	5/2004	Lee		
9,712,576	B1	7/2017	Gill	2004/0098291	A1	5/2004	Newburn		
9,785,445	B2	10/2017	Mitsui	2004/0125150	A1	7/2004	Adcock		
9,830,398	B2	11/2017	Schneider	2004/0162833	A1	8/2004	Jones		
9,842,312	B1	12/2017	Rosati	2004/0187089	A1	9/2004	Schulz		
9,949,681	B2	4/2018	Badenes	2004/0207249	A1	10/2004	Baumgartner		
9,953,282	B2	4/2018	Shaouy	2004/0230447	A1	11/2004	Schwerin-Wenzel		
9,959,420	B2	5/2018	Kiang	2004/0261013	A1	12/2004	Wynn		
9,978,040	B2	5/2018	Lee	2004/0268451	A1	12/2004	Robbin		
9,990,636	B1	6/2018	Lewis	2005/0144022	A1	6/2005	Evans		
10,001,911	B2	6/2018	Breedvelt-Schouten	2005/0210394	A1	9/2005	Crandall		
10,003,693	B2	6/2018	Wolthuis	2005/0216111	A1	9/2005	Nobuhiro		
10,083,412	B2	9/2018	Suntinger	2005/0222971	A1	10/2005	Cary		
10,157,355	B2	12/2018	Johnson	2006/0028917	A1	2/2006	Wigginton		
10,192,181	B2	1/2019	Katkar	2006/0047454	A1	3/2006	Tamaki		
10,235,156	B2	3/2019	Johnson	2006/0048151	A1	3/2006	Haruta		
10,264,067	B2	4/2019	Subramani	2006/0085245	A1	4/2006	Takatsuka		
10,308,992	B2	6/2019	Chauvin	2006/0095859	A1	5/2006	Bocking		
10,373,084	B2	8/2019	Kurjanowicz	2006/0136441	A1	6/2006	Fujisaki		
10,373,090	B2	8/2019	Holm	2006/0143270	A1	6/2006	Wodtke		
10,382,501	B2	8/2019	Malatesha	2006/0167736	A1	7/2006	Weiss		
10,455,011	B2	10/2019	Kendall	2006/0190391	A1	8/2006	Cullen		
10,496,943	B2	12/2019	De	2006/0200264	A1	9/2006	Kodama		
10,585,654	B2	3/2020	Singh	2006/0218551	A1	9/2006	Berstis		

(56)

References Cited

U.S. PATENT DOCUMENTS

2006/0224430	A1	10/2006	Butt		2010/0169146	A1	7/2010	Hoyne
2006/0235985	A1 *	10/2006	Ramkumar	 H04L 67/12 709/229	2010/0169802	A1	7/2010	Goldstein
2006/0277487	A1	12/2006	Poulsen		2010/0180212	A1	7/2010	Gingras
2007/0016646	A1	1/2007	Tendjoukian		2010/0223575	A1	9/2010	Leukart
2007/0025567	A1	2/2007	Fehr		2010/0269049	A1	10/2010	Fearon
2007/0038494	A1	2/2007	Kreitzberg		2010/0299171	A1	11/2010	Lau
2007/0041542	A1	2/2007	Schramm		2010/0312605	A1	12/2010	Mitchell
2007/0050225	A1	3/2007	Leslie		2010/0313151	A1	12/2010	Wei
2007/0073575	A1	3/2007	Yomogida		2011/0015961	A1	1/2011	Chan
2007/0143169	A1	6/2007	Grant		2011/0022662	A1	1/2011	Barber-Mingo
2007/0147178	A1	6/2007	Masuda		2011/0054968	A1	3/2011	Galaviz
2007/0150327	A1	6/2007	Dromgold		2011/0055177	A1	3/2011	Chakra
2007/0232278	A1	10/2007	May		2011/0060720	A1	3/2011	Devereux
2007/0255674	A1	11/2007	Mahoney		2011/0071878	A1	3/2011	Gingras
2007/0255715	A1	11/2007	Li		2011/0071893	A1	3/2011	Malhotra
2007/0260499	A1	11/2007	Greef		2011/0072372	A1	3/2011	Fritzley
2007/0288283	A1	12/2007	Fitzpatrick		2011/0093538	A1	4/2011	Weir
2007/0294344	A1	12/2007	Mohan		2011/0093619	A1	4/2011	Nelson
2008/0033777	A1	2/2008	Shukoor		2011/0113365	A1	5/2011	Kimmerly
2008/0046471	A1	2/2008	Moore		2011/0154216	A1	6/2011	Aritsuka
2008/0079730	A1	4/2008	Zhang		2011/0161128	A1	6/2011	Barney
2008/0082389	A1	4/2008	Gura		2011/0184768	A1	7/2011	Norton
2008/0082956	A1	4/2008	Gura		2011/0238706	A1	9/2011	Wong
2008/0091782	A1	4/2008	Jakobson		2011/0270644	A1	11/2011	Roncolato
2008/0120129	A1	5/2008	Seubert		2011/0302521	A1	12/2011	Jiang
2008/0126930	A1	5/2008	Scott		2011/0307100	A1	12/2011	Schmidtke
2008/0133524	A1	6/2008	Ryan		2011/0307772	A1	12/2011	Lloyd
2008/0134069	A1	6/2008	Horvitz		2012/0030194	A1	2/2012	Jain
2008/0155547	A1	6/2008	Weber		2012/0035942	A1	2/2012	Graupner
2008/0158023	A1	7/2008	Chung		2012/0066030	A1	3/2012	Limpert
2008/0167937	A1	7/2008	Coughlin		2012/0066411	A1	3/2012	Jeide
2008/0175104	A1	7/2008	Grieb		2012/0072251	A1	3/2012	Mircean
2008/0195964	A1	8/2008	Randell		2012/0079449	A1	3/2012	Sanderson
2008/0221946	A1	9/2008	Balon		2012/0110087	A1	5/2012	Culver
2008/0222566	A1	9/2008	Daughtrey		2012/0117499	A1	5/2012	Mori
2008/0244582	A1	10/2008	Brown		2012/0123835	A1	5/2012	Chu
2008/0268876	A1	10/2008	Gelfand		2012/0131191	A1	5/2012	May
2008/0270198	A1	10/2008	Graves		2012/0158757	A1	6/2012	Bhattacharjee
2008/0281665	A1	11/2008	Opaluch		2012/0158946	A1	6/2012	Shafiee
2008/0282326	A1 *	11/2008	Miller	 H04L 63/104 726/4	2012/0192086	A1	7/2012	Ghods
2008/0313004	A1	12/2008	Ryan		2012/0221963	A1	8/2012	Motoyama
2008/0313730	A1 *	12/2008	Ifimie	 H04L 63/20 726/17	2012/0239451	A1	9/2012	Caligor
2009/0048986	A1	2/2009	Anderson		2012/0254218	A1	10/2012	Ali
2009/0055796	A1	2/2009	Springborn		2012/0266068	A1	10/2012	Ryman
2009/0076878	A1	3/2009	Woerner		2012/0278388	A1	11/2012	Kleinbart
2009/0089133	A1	4/2009	Johnson		2012/0296993	A1	11/2012	Heyman
2009/0094623	A1	4/2009	Chakra		2012/0304187	A1	11/2012	Maresh
2009/0113310	A1	4/2009	Appleyard		2012/0317108	A1	12/2012	Okazaki
2009/0133027	A1	5/2009	Gunning		2013/0007332	A1	1/2013	Teh
2009/0167553	A1	7/2009	Hong		2013/0013560	A1	1/2013	Goldberg
2009/0187454	A1	7/2009	Khasin		2013/0014023	A1	1/2013	Lee
2009/0199192	A1	8/2009	Laithwaite		2013/0018688	A1	1/2013	Nudd
2009/0204463	A1	8/2009	Burnett		2013/0021629	A1	1/2013	Kurilin
2009/0204471	A1	8/2009	Elenbaas		2013/0066944	A1	3/2013	Laredo
2009/0234699	A1	9/2009	Steinglass		2013/0067375	A1	3/2013	Kim
2009/0241053	A1	9/2009	Augustine		2013/0067549	A1	3/2013	Caldwell
2009/0260010	A1	10/2009	Burkhart		2013/0073328	A1	3/2013	Ehrler
2009/0287523	A1	11/2009	Lau		2013/0103412	A1	4/2013	Nudd
2009/0296908	A1	12/2009	Lee		2013/0124638	A1	5/2013	Barreto
2009/0299803	A1	12/2009	Lakritz		2013/0151421	A1	6/2013	Van Der Ploeg
2009/0307319	A1	12/2009	Dholakia		2013/0151604	A1	6/2013	Ranade
2010/0005087	A1	1/2010	Basco		2013/0173486	A1	7/2013	Peters
2010/0070888	A1	3/2010	Watabe		2013/0179208	A1	7/2013	Chung
2010/0081417	A1 *	4/2010	Hickie	 G06F 21/604 455/414.1	2013/0179799	A1	7/2013	Savage
2010/0088137	A1	4/2010	Weiss		2013/0215116	A1	8/2013	Siddique
2010/0106627	A1	4/2010	O'Sullivan		2013/0227007	A1	8/2013	Savage
2010/0114786	A1	5/2010	Aboujaoude		2013/0246110	A1	9/2013	Nakhayi Ashtiani
2010/0115523	A1	5/2010	Kuschel		2013/0246399	A1	9/2013	Schneider
2010/0122334	A1	5/2010	Stanzione		2013/0275229	A1	10/2013	Moganti
2010/0131860	A1	5/2010	Dehaan		2013/0279685	A1	10/2013	Kohler
2010/0145801	A1	6/2010	Chekuri		2013/0317871	A1	11/2013	Kulkarni
					2013/0321467	A1	12/2013	Tappen
					2013/0339099	A1	12/2013	Aidroos
					2013/0339831	A1	12/2013	Gulanikar
					2014/0007005	A1	1/2014	Libin
					2014/0012603	A1	1/2014	Scanlon
					2014/0025767	A1	1/2014	De Kezel
					2014/0036639	A1	2/2014	Taber
					2014/0040780	A1	2/2014	Brian

US 12,401,655 B1

Page 4

(56)

References Cited

U.S. PATENT DOCUMENTS

2014/0040905	A1	2/2014	Tadanobu	2017/0017364	A1	1/2017	Kekki	
2014/0058801	A1	2/2014	Deodhar	2017/0017924	A1	1/2017	Kashiwagi	
2014/0059910	A1	3/2014	Norton	2017/0039503	A1	2/2017	Jones	
2014/0074536	A1	3/2014	Meushar	2017/0061341	A1	3/2017	Haas	
2014/0081685	A1	3/2014	Thacker	2017/0068933	A1	3/2017	Norton	
2014/0089719	A1	3/2014	Daum	2017/0093871	A1 *	3/2017	Abuelsaad	H04L 63/101
2014/0101310	A1	4/2014	Savage	2017/0093874	A1	3/2017	Uthe	
2014/0156539	A1	6/2014	Brunet	2017/0099296	A1	4/2017	Fisher	
2014/0165001	A1	6/2014	Shapiro	2017/0103369	A1	4/2017	Thompson	
2014/0172478	A1	6/2014	Vadasz	2017/0116552	A1	4/2017	Deodhar	
2014/0189017	A1	7/2014	Prakash	2017/0132200	A1	5/2017	Noland	
2014/0200944	A1	7/2014	Henriksen	2017/0153799	A1	6/2017	Hoyer	
2014/0208325	A1	7/2014	Chen	2017/0154024	A1	6/2017	Subramanya	
2014/0215344	A1	7/2014	Ligman	2017/0177671	A1	6/2017	Allgaier	
2014/0229609	A1	8/2014	Wong	2017/0185592	A1	6/2017	Frei	
2014/0236663	A1	8/2014	Smith	2017/0192642	A1	7/2017	Fishman	
2014/0244334	A1	8/2014	De	2017/0206217	A1	7/2017	Deshpande	
2014/0257894	A1	9/2014	Melahn	2017/0237745	A1 *	8/2017	Desai	H04L 63/104
2014/0279294	A1	9/2014	Field-Darragh					726/1
2014/0288987	A1	9/2014	Liu	2017/0249577	A1	8/2017	Nishikawa	
2014/0310047	A1	10/2014	De	2017/0316367	A1	11/2017	Candito	
2014/0310051	A1	10/2014	Meng	2017/0317898	A1	11/2017	Candito	
2014/0343989	A1 *	11/2014	Martini	2017/0323233	A1	11/2017	Bencke	
				2017/0323267	A1	11/2017	Baek	
				2017/0323350	A1	11/2017	Laderer	
				2017/0344754	A1	11/2017	Kumar	
				2017/0346861	A1	11/2017	Pearl	
				2017/0351385	A1	12/2017	Ertmann	
				2018/0032524	A1	2/2018	Byron	
				2018/0052943	A1	2/2018	Hui	
				2018/0053127	A1	2/2018	Boileau	
				2018/0059910	A1	3/2018	Wooten	
				2018/0060785	A1	3/2018	Carnevale	
				2018/0060818	A1	3/2018	Ishiyama	
				2018/0063063	A1	3/2018	Yan	
				2018/0068271	A1	3/2018	Abebe	
				2018/0075387	A1	3/2018	Kulkarni	
				2018/0088754	A1	3/2018	Psenka	
				2018/0089625	A1	3/2018	Rosati	
				2018/0095938	A1	4/2018	Monte	
				2018/0102989	A1	4/2018	Borsutsky	
				2018/0131649	A1	5/2018	Ma	
				2018/0157477	A1	6/2018	Johnson	
				2018/0165610	A1	6/2018	Dumant	
				2018/0173386	A1	6/2018	Adika	
				2018/0189706	A1	7/2018	Newhouse	
				2018/0189736	A1	7/2018	Guo	
				2018/0225795	A1	8/2018	Napoli	
				2018/0247352	A1	8/2018	Rogers	
				2018/0260081	A1	9/2018	Beaudoin	
				2018/0262620	A1	9/2018	Wolthuis	
				2018/0284985	A1	10/2018	Bansal	
				2018/0285471	A1	10/2018	Hao	
				2018/0316636	A1	11/2018	Kamat	
				2018/0331842	A1	11/2018	Faulkner	
				2018/0357049	A1	12/2018	Epstein	
				2018/0367477	A1	12/2018	Hariram	
				2018/0367483	A1	12/2018	Rodriguez	
				2018/0373804	A1	12/2018	Zhang	
				2019/0005048	A1	1/2019	Crivello	
				2019/0014070	A1	1/2019	Mertvetsov	
				2019/0018552	A1	1/2019	Bloy	
				2019/0034057	A1	1/2019	Rudchenko	
				2019/0068390	A1	2/2019	Gross	
				2019/0079909	A1	3/2019	Purandare	
				2019/0080289	A1	3/2019	Kreitler	
				2019/0095839	A1	3/2019	Yuki	
				2019/0095846	A1	3/2019	Gupta	
				2019/0102700	A1	4/2019	Babu	
				2019/0138583	A1	5/2019	Silk	
				2019/0138589	A1	5/2019	Udell	
				2019/0138961	A1	5/2019	Santiago	
				2019/0139004	A1	5/2019	Vukovic	
				2019/0147386	A1	5/2019	Balakrishna	
				2019/0187987	A1	6/2019	Fauchère	
				2019/0213509	A1	7/2019	Burleson	
				2019/0265821	A1	8/2019	Pearl	
				2019/0327271	A1 *	10/2019	Saxena	H04L 41/16
				2019/0340166	A1	11/2019	Raman	
2014/0040905	A1	2/2014	Tadanobu					
2014/0058801	A1	2/2014	Deodhar					
2014/0059910	A1	3/2014	Norton					
2014/0074536	A1	3/2014	Meushar					
2014/0081685	A1	3/2014	Thacker					
2014/0089719	A1	3/2014	Daum					
2014/0101310	A1	4/2014	Savage					
2014/0156539	A1	6/2014	Brunet					
2014/0165001	A1	6/2014	Shapiro					
2014/0172478	A1	6/2014	Vadasz					
2014/0189017	A1	7/2014	Prakash					
2014/0200944	A1	7/2014	Henriksen					
2014/0208325	A1	7/2014	Chen					
2014/0215344	A1	7/2014	Ligman					
2014/0229609	A1	8/2014	Wong					
2014/0236663	A1	8/2014	Smith					
2014/0244334	A1	8/2014	De					
2014/0257894	A1	9/2014	Melahn					
2014/0279294	A1	9/2014	Field-Darragh					
2014/0288987	A1	9/2014	Liu					
2014/0310047	A1	10/2014	De					
2014/0310051	A1	10/2014	Meng					
2014/0343989	A1 *	11/2014	Martini					
2014/0350997	A1	11/2014	Holm					
2014/0364987	A1	12/2014	Shikano					
2015/0006448	A1	1/2015	Gupta					
2015/0007058	A1	1/2015	Wooten					
2015/0012330	A1	1/2015	Sugiura					
2015/0052437	A1	2/2015	Crawford					
2015/0058053	A1	2/2015	De					
2015/0113540	A1	4/2015	Rabinovici					
2015/0134393	A1	5/2015	De					
2015/0153906	A1	6/2015	Liao					
2015/0154526	A1	6/2015	Yates					
2015/0178666	A1	6/2015	Green					
2015/0200966	A1 *	7/2015	Kasturirangan					
2015/0213411	A1	7/2015	Swanson					
2015/0215256	A1	7/2015	Ghafourifar					
2015/0262111	A1	9/2015	Yu					
2015/0312375	A1	10/2015	Valey					
2015/0317595	A1	11/2015	De					
2015/0339006	A1	11/2015	Chaland					
2015/0363092	A1	12/2015	Morton					
2015/0363733	A1	12/2015	Brown					
2015/0379472	A1	12/2015	Gilmour					
2016/0012368	A1	1/2016	O'Connell					
2016/0034844	A1	2/2016	Kofman					
2016/0048408	A1	2/2016	Madhu					
2016/0048786	A1	2/2016	Fukuda					
2016/0063192	A1	3/2016	Johnson					
2016/0063449	A1	3/2016	Duggan					
2016/0072750	A1	3/2016	Kass					
2016/0092045	A1	3/2016	Lamas					
2016/0110670	A1	4/2016	Chatterjee					
2016/0124775	A1	5/2016	Ashtiani					
2016/0140474	A1	5/2016	Vekker					
2016/0140501	A1	5/2016	Figlin					
2016/0147773	A1	5/2016	Smith					
2016/0147846	A1	5/2016	Smith					
2016/0148157	A1	5/2016	Walia					
2016/0180277	A1	6/2016	Skiba					
2016/0180298	A1	6/2016	McClement					
2016/0182311	A1	6/2016	Borna					
2016/0188145	A1	6/2016	Gabor					
2016/0216854	A1	7/2016	McClellan					
2016/0224939	A1	8/2016	Chen					
2016/0234391	A1	8/2016	Wolthuis					
2016/0275436	A1	9/2016	Kurjanowicz					
2016/0313934	A1	10/2016	Isherwood					
2016/0328217	A1	11/2016	Hagerty					
2016/0342927	A1	11/2016	Reznik					
2017/0004213	A1	1/2017	Cunico					
2017/0009387	A1	1/2017	Ge					

(56)

References Cited**U.S. PATENT DOCUMENTS**

2019/0340296	A1	11/2019	Cunico
2019/0340574	A1	11/2019	Ekambaram
2019/0347094	A1	11/2019	Sullivan
2019/0347126	A1	11/2019	Bhandari
2019/0370320	A1	12/2019	Kalra
2020/0019907	A1	1/2020	Notani
2020/0059539	A1	2/2020	Wang
2020/0065736	A1	2/2020	Relangi
2020/0162315	A1	5/2020	Siddiqi
2020/0192538	A1	6/2020	Karpe
2020/0192908	A1	6/2020	Smith
2020/0193556	A1	6/2020	Jin
2020/0218551	A1	7/2020	Sabo
2020/0228474	A1	7/2020	Cameron
2020/0233879	A1	7/2020	Papanicolaou
2020/0244611	A1	7/2020	Rosenstein
2020/0293975	A1	9/2020	Faulkner
2020/0328906	A1	10/2020	Raghavan
2020/0344253	A1	10/2020	Kurup
2021/0004380	A1	1/2021	Koch
2021/0004381	A1	1/2021	Smith
2021/0097466	A1	4/2021	Sabo
2021/0103451	A1	4/2021	Sabo
2021/0110347	A1	4/2021	Khalil
2021/0117907	A1	4/2021	Gray
2021/0136012	A1	5/2021	Barbitta
2021/0182475	A1	6/2021	Pelz
2021/0216562	A1	7/2021	Smith
2021/0232282	A1	7/2021	Karpe
2021/0248161	A1	8/2021	Leston
2021/0320891	A1	10/2021	Rosenstein
2021/0342786	A1	11/2021	Jiang
2021/0382734	A1	12/2021	Rosenstein
2022/0019320	A1	1/2022	Sabo
2022/0058548	A1	2/2022	Garg
2022/0075792	A1	3/2022	Koch
2022/0078142	A1	3/2022	Cameron
2022/0158859	A1	5/2022	Raghavan
2023/0161955	A1	5/2023	Pelz

FOREIGN PATENT DOCUMENTS

CN	102378975	B	5/2015
WO	2015036817	A1	3/2015
WO	2015123751	A1	8/2015
WO	2020006634	A1	1/2020

OTHER PUBLICATIONS

"U.S. Appl. No. 14/584,750, Non Final Office Action mailed Aug. 28, 2015", 21 pgs.

"U.S. Appl. No. 14/584,750, Notice of Allowance mailed Mar. 28, 2016", 8 pgs.

"U.S. Appl. No. 14/584,750, Response filed Feb. 29, 2015 to Non Final Office Action mailed Aug. 28, 2015", 16 pgs.

"U.S. Appl. No. 14/584,850, Final Office Action mailed Sep. 1, 2017", 31 pgs.

"U.S. Appl. No. 14/584,850, Non Final Office Action mailed Jan. 10, 2017", 9 pgs.

"U.S. Appl. No. 14/584,850, Response filed Apr. 10, 2017 to Non Final Office Action mailed Jan. 10, 2017", 13 pgs.

"How to Asana: Inviting teammates to Asana." YouTube, Asana, Mar. 21, 2017, <https://www.youtube.com/watch?v=TL0ruY1KyxU> (Year: 2017), 13 pages.

"Rules of Data Conversion from Document to Relational Data-bases", published: 2014, publisher: Future-processing, pp. 1-8 (Year: 2014).

(Tiburca, Andrew) Best Team Calendar Applications for 2018-Toggl <https://toggl.com/blog/best-team-calendar-applications-for-2018> (Year: 2017) 3 pages.

Asana Demo and Product Tour, you tube excerpt, Dec. 7, 2017 <https://www.youtube.com/watch?v=IMAFVVLGFyw> (Year: 2017) (16 pages).

Asana integrations, Asana tutorial, youtube, excerpt, Nov. 16, 2016 <https://www.youtube.com/watch?v=hBiQ7DJNinE> (Year: 2016) (21 pages).

Asana Workload and Portfolios, youtube, excerpt, Aug. 1, 2019 <https://www.youtube.com/watch?v=7XkNcfFDG6M> (Year: 2019) (20 pages).

Asana YouTube channel, list of all product videos, Nov. 19, 2014-Aug. 19, 2019 https://www.youtube.com/user/AsanaTeam/videos?disable_polymer=1 (Year: 2019) (5 pages).

Asana, Task dependencies, archives org, Aug. 25, 2017 <https://web.archive.org/web/20170825002141/https://asana.com/guide/help/tasks/dependencies> (Year: 2017) (5 pages).

Asana, Manage your team capacity with Workload, youtube, excerpt, Aug. 1, 2019 <https://www.youtube.com/watch?v=2ufXyZDzZnA&list=PLJFG93oi0wJAiUwyOhIGWHdtJzJrziIbV> (Year: 2019) (1 page).

Assef, F., Cassius, T. S., & Maria, T. S. (2018). Confrontation between techniques of time measurement. *Journal of Manufacturing Technology Management*, 29(5), 789-810. (Year: 2018).

Biggs, "GateGuru Relaunches With New Ways to Streamline Your Travel Experience", Techcrunch, (Apr. 26, 2013), 3 pgs.

Castaneda Samuel, Introduction Manual—Asana, Sep. 25, 2017 <https://static1.squarespace.com/static/586d532ae58c623db243a65/t/5c210c10f950b7fc7a8e3274/1545669658049/Asana+Manual.pdf> (Year: 2017) (20 pages).

Command and control, wikipedia, archives org, Mar. 16, 2018 https://web.archive.org/web/20180316193655/https://en.wikipedia.org/wiki/Command_and_control (Year: 2018), 6 pages.

Creating Tables with Fields from 2 Different Tables, published: 2009, publisher: StackOverflow, pp. 1-2. (Year: 2009).

Critical chain project management, Wikipedia, archives org, Dec. 17, 2016 https://web.archive.org/web/20161217090326/https://en.wikipedia.org/wiki/Critical_chain_project_management (Year: 2016) 5 pages.

Critical Path Method, Wikipedia, archives org, Sep. 19, 2017 https://web.archive.org/web/20170919223814/https://en.wikipedia.org/wiki/Critical_path_method (Year: 2017) 6 pages.

Dawei Li, "Deepcham: Collaborative Edge-Mediated Adaptive Deep Learning for Mobile Object Recognition", 2016, IEEE/ACM, pp. 64-76. (Year: 2016).

Devon Watts and Nick Fassler, "Now rolling out in beta: Track anything in Asana, with custom fields", publisher: Asana, published: Jul. 12, 2016, pp. 1-7 (Year: 2016).

Fang Chen, R. O. Briggs, "Project Progress Tracking Template - Using a Repeatable GSS Process to Facilitate Project Process Management," Proceedings of the 39th Annual Hawaii International Conference on System Sciences (HICSS'06), Kauai, HI, USA, 2006, pp. 17c-17c, doi: 10.1109/HICSS.2006.399. (Year: 2006).

Fruhlinger, Joshua. "The Best To-Do ListApps for Feeling Productive; With the right app, feeling productive can be just as gratifying as actually getting things done" Wall Street Journal (Online); New York, N.Y. [New York, N.Y.] Nov. 8, 2013 (Year: 2013) 4 pages.

Hartmann, "TimeProjectscheduling with resource capacities and requests varying with time: a case study," 2013, Flexible services and manufacturing journal, vol. 25, No. 1, pp. 74-93 (Year: 2013).

Helen Mongan-Rallis & Terrie Shannon, "Synchronous Chat," Aug. 2016, Dept. of Education, Univ. of MN Duluth, web.archive.org/web/20160825183503/https://www.d.umn.edu/hrallis/professional/presentations/cotfsp06/indiv_tools/sync_chat.htm (Year: 2016) (2 pages).

How to Asana Asana time tracking, youtube, excerpt, May 24, 2017 <https://www.youtube.com/watch?v=z91qlex-TLc> (Year: 2017) (1 page).

How to Asana, Asana project management, youtube, excerpt, Mar. 7, 2017 <https://www.youtube.com/watch?v=qqANMTvVpE> (Year: 2017) (28 pages).

How to Asana, Creating your first Asana project, youtube, excerpt, Jan. 31, 2017 <https://www.youtube.com/watch?v=L04WmcUdsLo> (Year: 2017) (1 page).

How to Asana, Getting Asana into your workflow, youtube, excerpt, Jul. 17, 2017 <https://www.youtube.com/watch?v=7YLRnMdv30> (Year: 2017) (24 pages).

(56)

References Cited**OTHER PUBLICATIONS**

How to Asana, Planning with Asana calendar, youtube excerpt, Feb. 14, 2017 <https://www.youtube.com/watch?v=w816KYiVPyc> (Year: 2017) (19 pages).

How to Asana, Using Asana for task management, youtube, excerpt, Feb. 7, 2017 <https://www.youtube.com/watch?v=vwvbgiejhQ> (Year: 2017) (8 pages).

How to Asana, Visualizing work with Asana kanban boards, youtube, excerpt, Feb. 21, 2017 <https://www.youtube.com/watch?v=jmZaZGydfPY> (Year: 2017) (41 pages).

How to Asana, Workflow management, youtube, excerpt, May 30, 2017 <https://www.youtube.com/watch?v=rk8nPwMxsRo> (Year: 2017) (9 pages).

How to use Advanced Search in Asana, Asana tutorial, May 25, 2016 <https://www.youtube.com/watch?v=5VyJ3toPfQM> (Year: 2016) (28 pages).

Justin Rosenstein, Unveiling the Future of Asana, Mar. 28, 2018 https://www.youtube.com/watch?v=nRI?d_WM4Bc (Year: 2018) (2 pages).

Auren Labrecque, "Fostering Consumer-Brand Relationships in Social Media Environments: The Role of Parasocial Interaction", 2014, Journal of Interactive Marketing, 28 (2014), pp. 134-148 (Year: 2014).

M. Ramzan, M. A. Iqbal, M. A. Jaffar, A. Rauf, S. Anwar and A. A. Shahid, "Project Scheduling Conflict Identification and Resolution Using Genetic Algorithms," 2010 International Conference on Information Science and Applications, Seoul, Korea (South), 2010, pp. 1-6, doi: 10.1109/ICISA.2010.5480400. (Year: 2010).

Macro, computer science, wikipedia, archives org, 6 pages, Feb. 11, 2020 [http://web.archive.org/web/20200211082902/https://en.wikipedia.org/wiki/Macro_\(computer_science\)](http://web.archive.org/web/20200211082902/https://en.wikipedia.org/wiki/Macro_(computer_science)) (Year: 2020) 6 pages.

Mauricio Aizawa, Zapier, How to Automate Asana Tasks creation using Evernote, youtube excerpts, Mar. 16, 2018 <https://www.youtube.com/watch?v=BjDQ4Gny4WI> (Year: 2018) (8 pages).

Paul Minors, How to automate your tasks, youtube excerpts, Oct. 18, 2019 <https://www.youtube.com/watch?v=IwF9XyUQrzw> (Year: 2019).

Prioritize My Tasks in Asana, Asana tutorial, youtube, excerpt, May 25, 2016 <https://www.youtube.com/watch?v=UbCnMvw01nl> (Year: 2016) (3 pages).

Project views, Asana tutorial, youtube, excerpt May 25, 2016 <https://www.youtube.com/watch?v=FYjA8ZH3ceQ> (Year: 2016) (5 pages).

Ruoming Pang, Ramon Caceres, Mike Burrows, Zhifeng Chen, Pratik Dave, Nathan Germer, Alexander Golynski, Kevin Graney, Nina Kang, Lea Kissner, Jeffrey L. Korn, Abhishek Parmar, Christina D. Richards, and Mengzhi Wang, "Zanzibar: Google's Consistent, Global Authorization System", 2019 USENIX Annual Technical Conference (USENIX ATC '19), Renton, WA, pp. 1-14 (Year: 2019).

Szabolcs de Gyurky; Mark A. Tarbell, "Project and Task Organization," in The Cognitive Dynamics of Computer Science: Cost-Effective Large Scale Software Development IEEE, 2006, pp. 54-79, doi: 1002/0470036443.ch (Year: 2006).

Using Asana Premium, Asana tutorial, youtube, excerpt, Sep. 10, 2016 <https://www.youtube.com/watch?v=vMgLtDDmyeo> (Year: 2016) (4 pages).

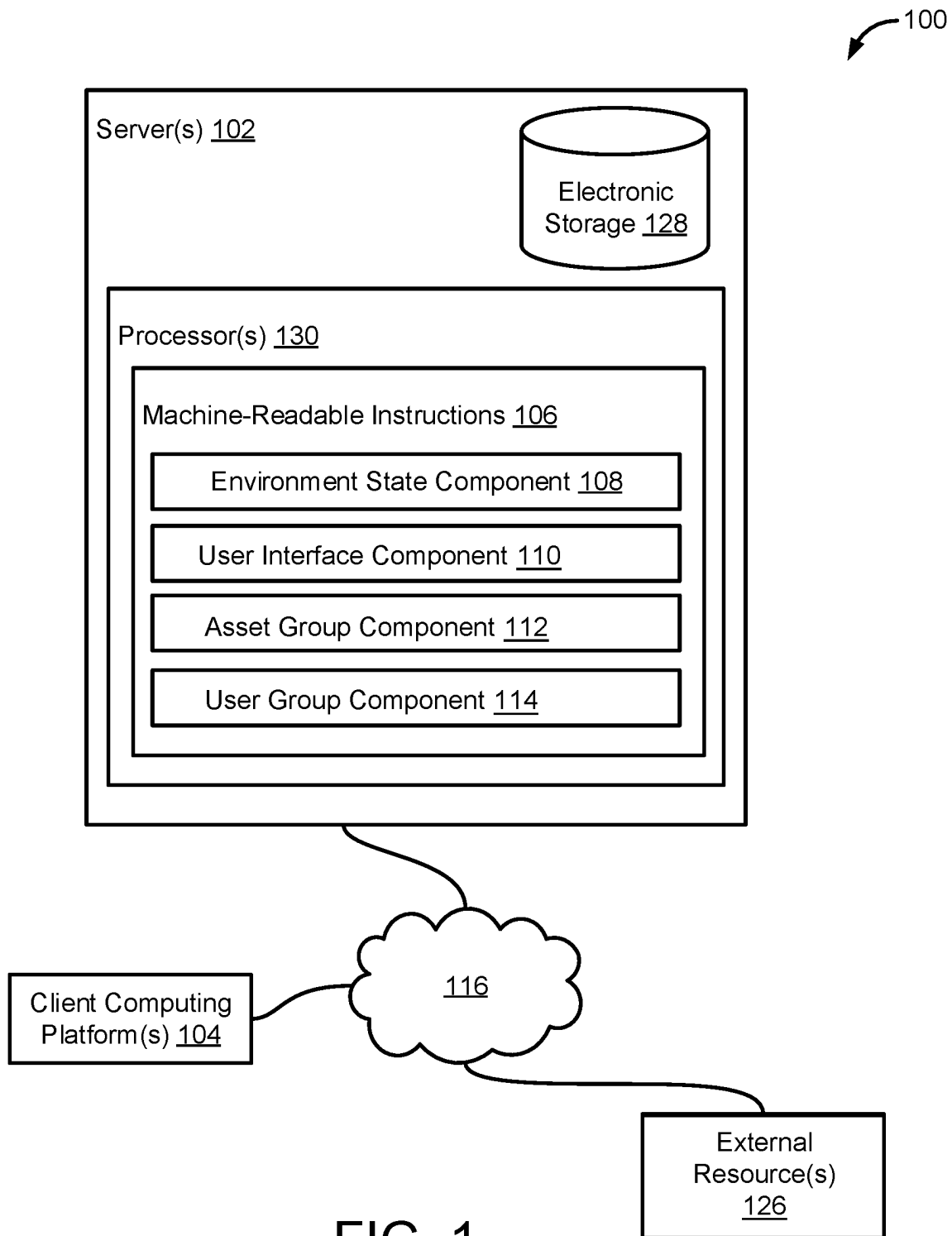
Where does Asana fit in, archives org, Jul. 8, 2017 <https://web.archive.org/web/20170708150928/https://asana.com/guide/resources/infosheets/where-does-asana-fit> (Year: 2017) (5 pages).

Wix.com, How to Use Wix Code with Marketing Tools to Create Custom Events, Oct. 18, 2018, YouTube, https://www.youtube.com/watch?v=MTBVykOYGvO&feature=emb_title, 2 pages.

www.asana.com (as retrieved from <https://web.archive.org/web/20160101054536/https://asana.com/press> and <https://web.archive.org/web/20160101054527/https://asana.com/product>) (Year: 2016) 15 pages.

www.cogmotive.com/blog/author/alan Alan Byrne: "Creating a company Shared Calendar in Office 365"; pp. 1-17; Sep. 10, 2013 (16 pages).

* cited by examiner



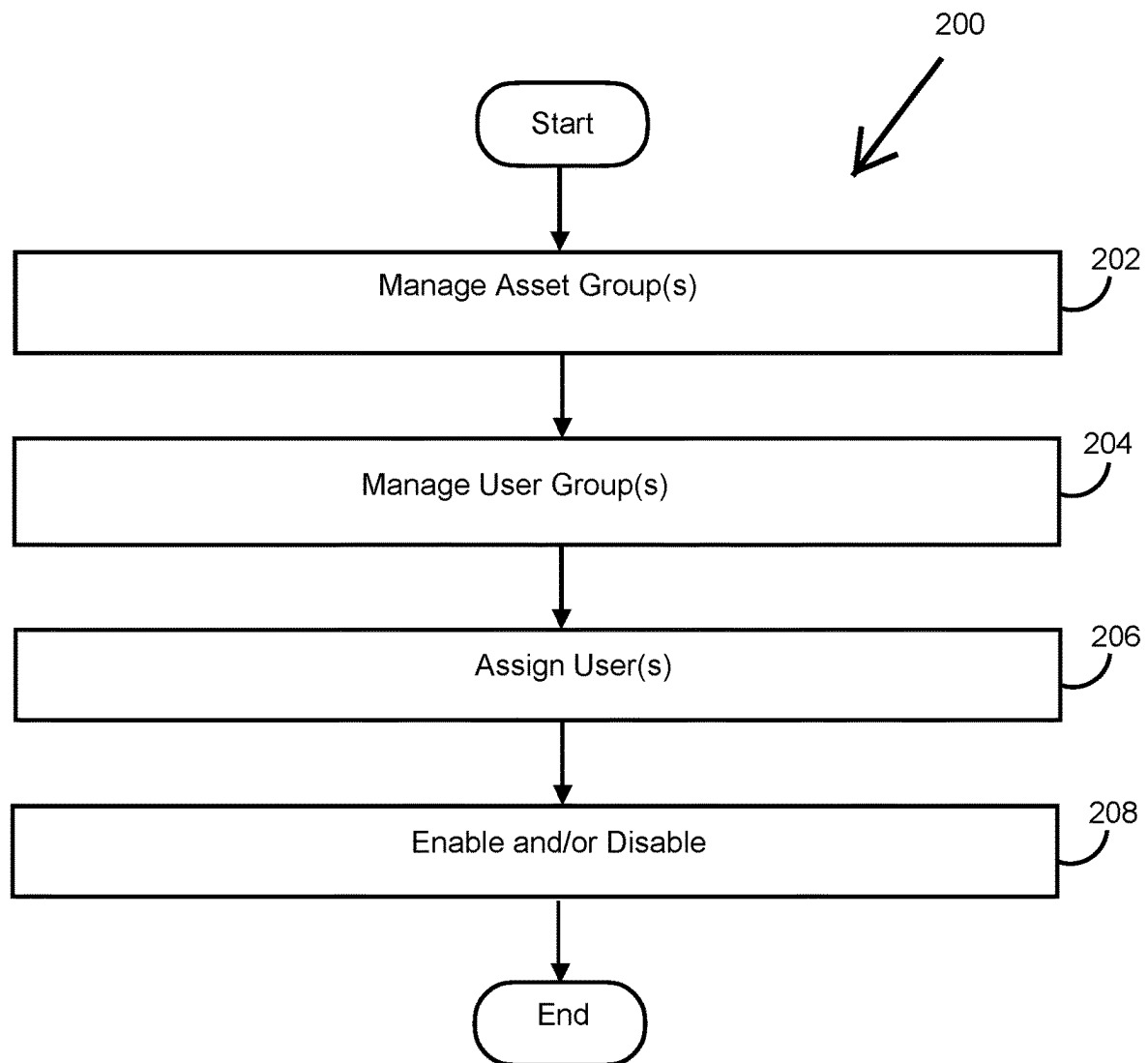


FIG. 2

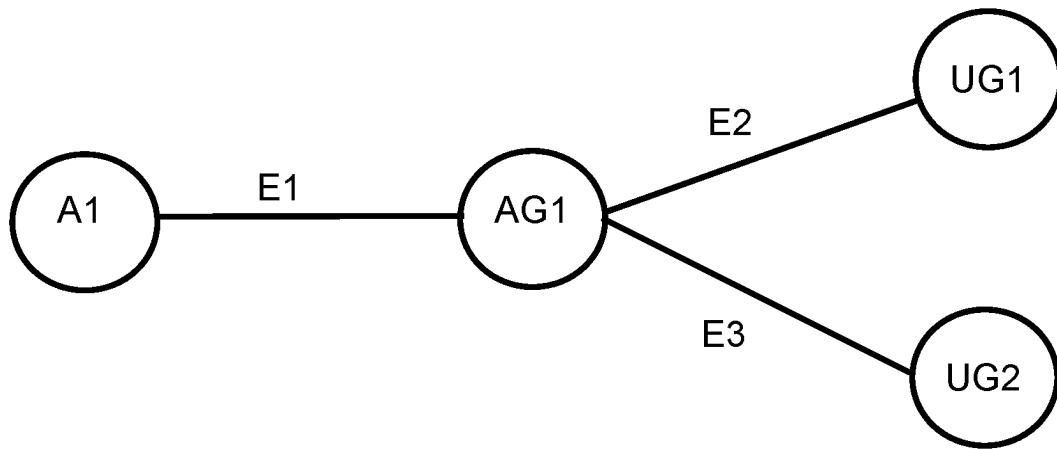


FIG. 3

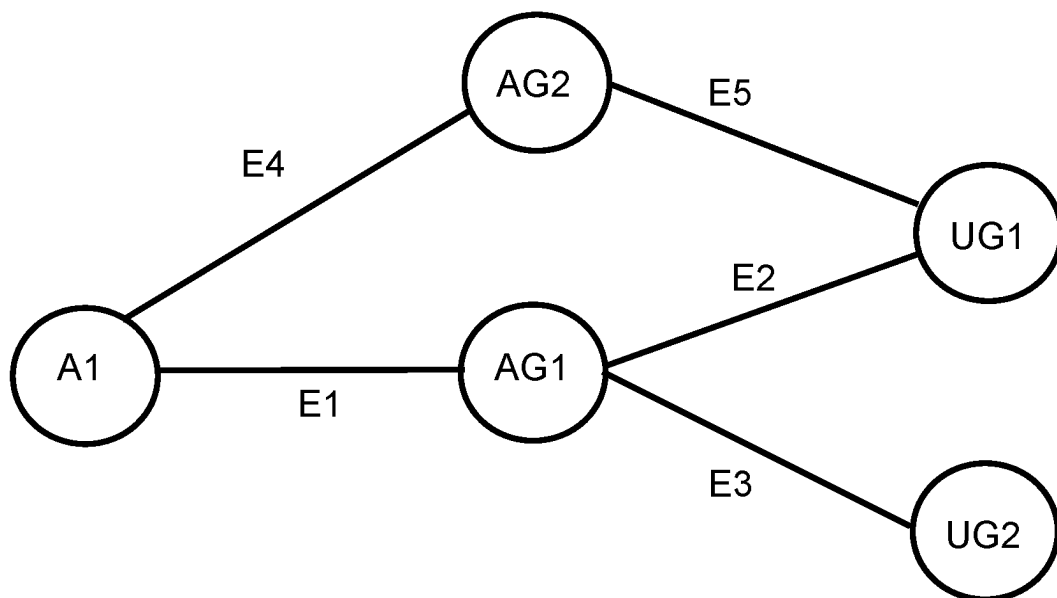


FIG. 4

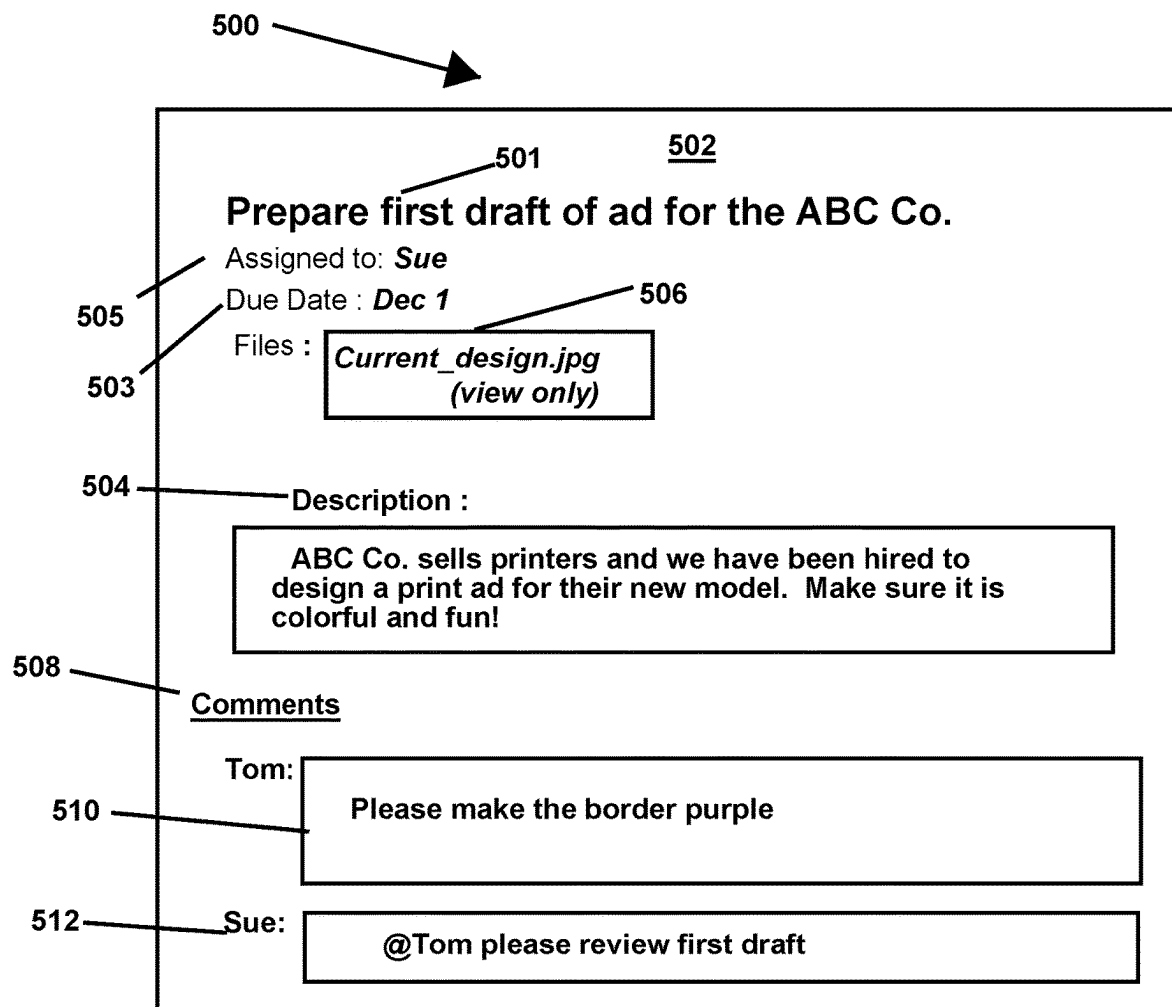


FIG. 5

1

SYSTEMS AND METHODS TO MANAGE ACCESS TO ASSETS OF A COMPUTER ENVIRONMENT BASED ON USER AND ASSET GROUPING

FIELD OF THE DISCLOSURE

The present disclosure relates to systems and methods to manage access to assets of a computer environment based on user and asset grouping.

BACKGROUND

Web-based computer environments, such as software-as-a-service (SaaS) products, may rely on a container- or folder-based data models to build and/or provide functionality to end users. By way of non-limiting illustration, assets making up a SaaS computer environment may only belong to a single container or hierarchy at a time. Conventional SaaS features may leverage the exclusive container/hierarchy configuration to build and/or organize functionality discretely. Some examples of functionality organized via these models may include, and/or may be related to, roles & permissions (authorization) assigned to users, licensing agreements granted to users, admin controls available to users, and/or control features available to users.

SUMMARY

Not all web-based computer environments may have a data model that is suitable to build out functionality in the same as or similar manner as conventional SaaS products. For example, a computer environment may define assets such that they belong to a single data object, no data object, within a hierarchy of data objects, and/or within multiple hierarchies simultaneously. It may be desired to replicate conventional SaaS behavior regardless of the hierarchy, or lack thereof. Thus, the inventors of the present disclosure have identified a need to configure and/or organize computer environment assets in a way that allows for discrete building and/or organization of functionality.

An example computer environment that may take advantage of the features and/or functionality of the present disclosure may include a collaboration environment. A collaboration environment, sometimes referred to a work management platform and/or other names, may enable users to assign projects, tasks, or other assignments to assignees (e.g., other users) to complete. A collaboration environment may comprise an environment in which individual users and/or a virtual team of users does its work and enables the users to work in a more organized and efficient manner when remotely located from each other.

A collaboration environment may be comprised of assets in the form of work unit records and/or other records for work. The records may be configured such that they are associated with zero, one, or many records within a hierarchy. Records may also exist in multiple hierarchies simultaneously. For example, a task record may be associated with multiple project records, with each of the project records being associated with multiple portfolio records. Because the records may be organized as such, replicating conventional SaaS behavior cannot be done through convention techniques. Thus, there is an unmet need to replicate conventional SaaS behavior regardless of the association of records with one or more other records and/or hierarchies.

To address these and/or other problems, one or more implementations presented herein propose a technique to

2

manage access to assets of a computer environment based on unique user and asset grouping.

One or more implementations of a system configured to manage access to assets of a computer environment based on user and asset grouping may include one or more hardware processors configured by machine-readable instructions and/or other components. Executing the machine-readable instructions may cause the one or more hardware processors to facilitate one or more features and/or functions presented herein. The machine-readable instructions may include one or more computer program components. The one or more computer program components may include one or more of an environment state component, an asset group component, a user group, and/or other components.

The environment state component may be configured to manage environment state information maintaining a computer environment. In some implementations, a computer environment may refer to one or more application programs running on a computer. The computer environment may be configured to facilitate interaction by users with the computer environment and/or each other through the computer environment. The environment state information may include information defining a computer environment through one or more of one or more user interfaces of the computer environment, the assets of the computer environment, access characteristics of the assets of the computer environment, and/or other aspects of the computer environment. By way of non-limiting illustration, the computer environment may include one or more of a collaboration environment, a customer relationship management ("CRM") program, a billing application, and/or other applications.

The asset group component may be configured to manage asset groups of a computer environment. The individual asset groups may be associated with one or more assets of a computer environment. The individual asset groups may define access characteristics of associated ones of the assets. By way of non-limiting illustration, the asset groups of a computer environment may include a first asset group and/or other asset groups. The first asset group may be associated with a first asset and may define a first set of access characteristics of the first asset of the computer environment.

The user group component may be configured to manage user groups. The individual user groups may be associated with one or more asset groups. The individual user groups may define the extent of the access characteristics defined by the individual asset groups that are granted to users through the associated individual user groups. By way of non-limiting illustration, the user groups may include one or more of a first user group, a second user group, and/or other user groups. The first user group may be associated with the first asset group and may define a first extent of the first set of access characteristics of the first asset that are granted to the users through the first user group. The second user group may be associated with the first asset group and may define a second extent of the first set of access characteristics of the first asset that are granted to the users through the second user group.

The environment state component may be configured to assign the users to the user groups. The assignment to the user groups may cause one or more of the access characteristics of associated ones of the assets to be granted to the users based on the extent of the access characteristics that are granted to the users through the user groups. By way of non-limiting illustration, a first user may be assigned to the first user group and/or other user groups; a second user may

3

be assigned to the second user group and/or other groups; and/or one or more other users may be assigned to one or more user groups.

The environment state component may be configured to enable and/or disable individual access characteristics of individual assets for individual users. Enabling and/or disabling individual access characteristics may be based on assigned ones of the user groups and the extent that the access characteristics defined by the asset groups are granted to the users through the user groups via the asset groups. By way of non-limiting illustration, the first extent of the first set of access characteristics may be enabled for the first user accessing the first asset of the computer environment based on the first user being assigned to the first user group. By way of non-limiting illustration, the second extent of the first set of access characteristics may be enabled for the second user accessing the first asset based on the second user being assigned to the second user group.

As used herein, any association (or relation, or reflection, or indication, or correspondency) involving servers, processors, client computing platforms, and/or another entity or object that interacts with any part of the system and/or plays a part in the operation of the system, may be a one-to-one association, a one-to-many association, a many-to-one association, and/or a many-to-many association or N-to-M association (note that N and M may be different numbers greater than 1).

These and other features, and characteristics of the present technology, as well as the methods of operation and functions of the related elements of structure and the combination of parts and economies of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the invention. As used in the specification and in the claims, the singular form of "a," "an," and "the" include plural referents unless the context clearly dictates otherwise.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a system configured to manage access to assets of a computer environment based on user and asset grouping, in accordance with one or more implementations.

FIG. 2 illustrates a method to manage access to assets of a computer environment based on user and asset grouping, in accordance with one or more implementations.

FIG. 3 illustrates relationships between user groups, an asset group, and an asset, in accordance with one or more implementations.

FIG. 4 illustrates relationships between user groups, asset groups, and an asset that may pose a conflict, in accordance with one or more implementations.

FIG. 5 illustrates a user interface of a collaboration environment, in accordance with one or more implementations.

DETAILED DESCRIPTION

FIG. 1 illustrates a system **100** configured to manage access to assets of a computer environment based on user and asset grouping, in accordance with one or more implementations. The system **100** may allow computer environ-

4

ments that do not have conventional SaaS data models to replicate conventional SaaS behavior to build and/or organize their functionally. One or more implementations of the system **100** may be particularly suited to handle access conflicts that may arise due to the non-SaaS data model configuration of such computer environments.

In some implementations, system **100** may include one or more of one or more servers **102**, one or more client computing platforms **104**, external resource(s) **126**, and/or other components. Server(s) **102** may be configured to communicate with one or more client computing platforms **104**, one or more external resources **126**, and/or other entities of system **100** according to a client/server architecture and/or other architectures. Client computing platform(s) **104** may be configured to communicate with other client computing platforms via server(s) **102** and/or according to a peer-to-peer architecture and/or other architectures. Users may access system **100** and/or instances of the collaboration environment via client computing platform(s) **104**. Server(s) **102** may be remote from client computing platform(s) **104**. Client computing platform(s) **104** may be remote from each other.

Server(s) **102** may include one or more of non-transitory electronic storage **128**, one or more processors **130** configured by machine-readable instructions **106**, and/or other components. The non-transitory electronic storage **128** may store environment state information and/or other information. Machine-readable instructions **106** may include one or more instruction components. The instruction components may include computer program components. Executing the machine-readable instructions **106** may cause server(s) **102** to facilitate managing access to assets of a computer environment based on user and asset grouping. The computer program components may include one or more of an environment state component **108**, a user interface component **110**, an asset group component **112**, a user group component **114**, and/or other components.

Environment state component **108** may be configured to manage environment state information maintaining a computer environment. In some implementations, a computer environment may refer to one or more application programs running on a computer. The computer environment may be configured to facilitate interaction by users with the computer environment and/or each other through the computer environment. The environment state information may include information defining a computer environment through one or more of user interface(s) of the computer environment, assets of the computer environment, access characteristics of the assets of the computer environment, and/or other aspects of the computer environment. By way of non-limiting illustration, the computer environment may include one or more of a collaboration environment, a customer relationship management ("CRM") program, a billing application, and/or other applications.

Assets of a computer environment may refer to information objects (sometimes referred to as "data objects" or simply "objects") that make up the computer environment, and/or other expressions of digitally stored information. A data object may comprise a region of storage that contains a value or group of values of one or more parameters. An individual value may be accessed using its identifier and/or a more complex expression that refers to an object. In some implementations, an object may have a unique data type.

In some implementations, access characteristics may define a user's ability to access, view, modify, and/or otherwise interact with one or more information objects of a computer environment through one or more user interfaces

5

of the computer environment. In some implementations, access characteristics granted to the users may be expressed as one or more of user roles within a computer environment, permissions the users have within a computer environment, user license agreements within the computer environment, control levels available to be exercised within a computer environment, and/or other access characteristics. By way of non-limiting illustration, different level accounts or licenses may be reflected by different types or levels of access to assets. By way of non-limiting illustration, users of a given role may have a certain level of control over one or more assets, while users of another role may have a different level of control of the one or more assets.

Environment state component 108 may be configured to manage environment state information maintaining a collaboration environment. The collaboration environment may be configured to facilitate interaction by users with the collaboration environment and/or each other. The environment state information may include one or more records. The records may include one or more of user records, work unit records, and/or other records. In some implementations, individual records and/or groups of records may from the assets of a collaboration environment. In some implementations, a calendar provided through the collaboration environment, or another application, may comprise an asset. By way of non-limiting illustration, an individual record may be in the form of an individual information object. The user records may include user information describing the users of the collaboration environment. The work unit records may include work unit information describing units of work assigned to, created by, and/or managed by the users within the collaboration environment. Although one or more implementations of the features and/or functions presented herein may refer to a collaboration environment, this is for illustrative purposes only and is not to be considered limiting. Instead, those skilled in the art will understand that such features and/or functions may similarly apply to other computer environments.

In some implementations, sets of the work unit records may be specifically designated as being associated with one or more record hierarchies. Individual work unit records associated with an individual hierarchy may be designated as being subordinate and/or superior to one or more other records within the individual hierarchy. These work unit records may have features and/or functionality specific to them, which are particularly well adapted for organizing, managing, and/or otherwise aiding in the ongoing nature of hierarchical definitions of work.

The user information in the user records may include values of user parameters and/or other information. The values of the user parameters may be organized in the user records corresponding to users interacting with and/or viewing the collaboration environment. The values of the user parameters may include information describing and/or identifying the users, their actions within the collaboration environment, their settings, metadata associated with the users, work they are members of and/or collaborate on, and/or other user information. Individual ones of the users may be associated with individual ones of the user records. A user record may define values of the user parameters associated with a given user.

The values of the user parameters may, by way of non-limiting example, specify one or more of: a user name, a group, a user account level, user role information, a user department, descriptive user content, a to-email, a from-email, a photo, an organization, a workspace, one or more user comments, one or more teams the user belongs to, one

6

or more of the user display settings (e.g., colors, size, project order, task order, other unit of work order, etc.), work inclusion information (e.g., identification of work unit records they are members of and/or collaborate on), one or more authorized applications, one or more interaction parameters (e.g., indicating a user is working on/worked on a given unit of work, a given user viewed a given work unit of work, a given user selected a given unit of work, a timeframe a given user last interacted with and/or worked on a given unit of work, a time period that a given unit of work has been idle, and/or other interaction parameters), one or more notification settings, one or more progress parameters, status information for one or more work units the user is associated with (units of work assigned to the user, assigned to other users by the user, completed by the user, past-due date, and/or other information), one or more performance/productivity metrics of a given user (e.g., how many units of work the user has completed, how quickly the user completed the units of work, how quickly the user completes certain types of work units, the efficiency of the user, bandwidth of the user, activity level of the user, how many business objectives the user has helped fulfill through their completion of units of work, etc.), application access information (e.g., username/password for one or more third-party applications), one or more favorites and/or priorities, schedule information, and/or other information.

Schedule information for the individual users may include one or more calendar entries associated with the individual users. The individual calendar entries may be associated with individual start dates and individual end dates. In some implementations, schedule information may be stored locally within electronic storage 128 by virtue of features and/or functionality provided within a collaboration environment. By way of non-limiting illustration, a collaboration environment may have features and/or functionality of a calendar application configured to facilitate calendaring entries into a schedule. It is noted that schedule information may be determined through features and/or functionality provided by one or more external resources 126. By way of non-limiting illustration, external resources may include a calendar application which may be external to a collaboration environment. The collaboration environment may have permissions to access the external calendar application to determine and/or obtain schedule information.

Work inclusion information may include identification(s) of work unit records that individual users are members of and/or collaborate with one or more other users on.

The “members” of the work unit records may include specifically named users within the work unit records. Specifically naming users may be accomplished by linking user records for those users to the work unit records. Linking the user records to the work unit records may comprise including a referential notation of the user records in the work unit records. Linking the user records to the work unit records may comprise including copies or instances of the user records in the work unit records. In some implementations, users having a particular status with respect to a work unit record may be considered members. By way of non-limiting illustration, users who are designated to perform one or more actions to facilitate completion of the units of work (e.g., assignees, assignors, creators, managers, collaborators, and/or other users) may be considered members. In some implementations, “member” status may be a status that is separate and distinct from users who are activity performing one or more actions to facilitate completion of the units of work. In some implementations, “member” may refer to users who view, access, monitor, and/or otherwise

engage with units of work in ways that may not directly facilitate completion of the units of work.

The work unit information in the work unit records may include values of one or more work unit parameters and/or other information. The values of the work unit parameters may be organized in work unit records corresponding to units of work managed, created, and/or assigned within the collaboration environment. A given unit of work may have one or more assignees, owners, and/or collaborators working on the given work unit. Units of work may include one or more of tasks, projects, portfolios, objectives, and/or other units of work.

Work unit records may be created by a given user for the given user and/or created by the given user and assigned to one or more other users. Work unit records may be created by a given user for the given user and/or created by the given user and managed by one or more other users.

Individual units of work may include and/or may be associated with one or more digital content items. An individual unit of work may include an individual digital content item by virtue of the individual digital content item (and/or a copy or instance thereof) being accessible via a work unit page of the unit of work (e.g., via link, URL, a pointer, and/or other techniques to provide access). A digital content item may include one or more of an image, a video, an audio file, a PDF, a word document, multimedia presentations, a webpage, and/or other digital content items.

In some implementations, work unit records created by, assigned to, and/or completed by the users may refer generally to a linking of the work unit records with the individual users in the collaboration environment. A work unit record may be linked with a user in a manner that defines one or more relationships between the user and the work unit record. Such a relationship may connote and/or be a result of an action (past, present, and/or future) of the user with respect to the work unit record and/or a unit of work. Such actions may include one or more of creating a work unit record for a unit of work, being assigned to participate in a unit of work, participating in a unit of work, being granted access to a work unit record of a unit of work, adjusting a value of a work unit parameter of a work unit record of a unit of work, being assigned a role at the unit of work level, and/or other actions.

In some implementations, values of work unit parameters may specify one or more of a unit of work name, a unit of work description, user role information, one or more unit of work dates (e.g., a start date, a due date or end date, a completion date, and/or other dates), monetary budget and/or consumption, an amount of work production, project inclusion (e.g., identification of projects supported by the individual units of work), objective inclusion (e.g., identification of objectives supported by the individual units of work), one or more members associated with a unit of work (e.g., an owner, one or more collaborators, collaborator access information, and/or other unit of work collaborators and/or collaborator information), completion state, one or more user comment parameters (e.g., permission for who may make comments such as an assignee, an assignor, a recipient, one or more followers, and/or one or more other interested parties; content of the comments; one or more times; presence or absence of the functionality of up-votes; one or more hard-coded responses; and/or other parameters), one or more interaction parameters (e.g., indicating a given unit of work is being worked on/was worked on, a given work unit of work was viewed, a given unit of work was selected, how long the given unit of work has been idle, a last interaction parameter indicating when and what user last

interacted with the given unit of work, users that interacted with the given unit of work, quantity and/or content of comments on the unit of work, and/or other interaction parameters indicating sources of the interactions, context of the interactions, content of the interactions and/or time for the interactions), one or more digital content item attachments, notification settings, privacy, an associated URL, one or more interaction parameters (e.g., sources of the interactions, context of the interactions, content of the interactions, time for the interactions, and/or other interaction parameters), updates, state of a workspace for a given unit of work (e.g., application state parameters, application status, application interactions, user information, and/or other parameters related to the state of the workspace for a unit of work), one or more performance/productivity metrics for a given unit of work, hierarchical information, dependency information, one or more custom fields (e.g., priority, cost, stage, and/or other custom fields), and/or other information.

In some implementations, work unit parameters may include one or more of a work assignment parameter, a work completion parameter, a work management parameter, a work creation parameter, a dependency parameter, a grouping parameter, and/or other parameters. The values of the work assignment parameter may describe assignees of individual units of work. The values of the work management parameter may describe users who manage individual units of work and/or the extent in which they manage. The values of the work creation parameter may describe creation characteristics of individual units of work. The creation characteristics may include who created the work unit record, when it was created, and/or other information.

In some implementations, values of a dependency parameter may describe whether a given unit of work is dependent on one or more other units of work. A unit of work being dependent on an other unit of work may mean the unit of work may not be completed, started, assigned, and/or have other interactions performed in relation to the unit of work before some action is performed on the other unit of work. By way of non-limiting illustration, a unit of work may not be started until another unit of work is completed, meaning the unit of work may be dependent on the other unit of work. In some implementations, values of the dependency parameter may go hand in hand with the hierarchical information. By way of non-limiting illustration, a unit of work that is subordinate to an other unit of work may be dependent on the other unit of work or vice versa.

The values of the work assignment parameter describing assignment of users to units of work may be determined based on one or more interactions by one or more users with a collaboration environment. In some implementations, one or more users may create and/or assign one or more unit of work to themselves and/or another user. In some implementations, a user may be assigned a unit of work and the user may effectuate a reassignment of the unit of work from the user or one or more other users.

In some implementations, values of the work completion parameter may indicate that a completion status of a unit of work has changed from "incomplete" to "marked complete" and/or "complete." In some implementations, a status of complete for a unit of work may be associated with the passing of an end date associated with the unit of work. In some implementations, a status of "marked complete" may be associated with a user providing input via the collaboration environment at the point in time the user completes the unit of work (which may be before or after an end date). In some implementations, a completion status may include one or more of "rejected," "revisions required," "incom-

plete,” and/or other statuses. By way of non-limiting illustration, for units of work that require one user to review work by another user, additional user interface elements may be presented where users can set the status as one or more of “marked complete,” “incomplete,” “rejected,” “revisions required,” and/or other statuses. In some implementations, a status of rejected for a unit of work may be associated with the passing of an end date associated with the unit of work without the work having been marked complete manually. In some implementations, a status of rejected for a unit of work may be associated with a user specifying the unit of work as rejected. In some implementations, a status of revisions required for a unit of work may be associated with a user specifying that the work being reviewed requires changes but is otherwise approved. In some implementations, a status of revisions required may present the reviewing user with options to specify what revisions are required. In some implementations, a status of revisions required may cause a follow up unit of work to be generated which is assigned back to the assignor of the original work.

In some implementations, values of a grouping parameter may describe whether a given unit of work supports (e.g., is included in) a project and/or objective. The values may specify one or more of a name/title of another work unit record (e.g., project, objective), a name/username of an owner of another work unit record, and/or other information.

In some implementations, work unit records may define tasks. The work unit records defining tasks may be referred to as task records. The work unit information in task records may be referred to as task information and/or other information. Task information may include values of work unit parameters for tasks managed within the collaboration environment. The work unit parameters for tasks may be referred to as “task parameters.” The work unit parameters comprising task parameters may be specific to tasks and/or may include one or more parameters not shared with projects, objectives, and/or other records. The task parameters may characterize one or more tasks created, assigned, and/or managed within the collaboration environment and/or via the collaboration environment and/or the metadata associated with the one or more tasks.

In some implementations, work unit records may define projects. The work unit records defining projects may be referred to as project records. The work unit information in project records may be referred to as project information and/or other information. Project information may include values of work unit parameters for projects managed within the collaboration environment. The work unit parameters for projects may be referred to as “project parameters.” The work unit parameters comprising project parameters may be specific to project records and may include one or more parameters not shared with tasks, objectives, and/or other records. The project parameters may characterize one or more projects created, owned, assigned, and/or managed within the collaboration environment and/or via the collaboration environment and/or the metadata associated with the one or more projects. Individual ones of the projects may be associated with individual ones of the records. A given project may have one or more owners and/or one or more collaborators working on the given project. The given project may be associated with one or more other units of work assigned to one or more users under the given project heading. In some implementations, projects may include one or more units of work that may directly facilitate progress toward fulfillment of the projects. Accordingly, completion of units of work may directly contribute to progress toward fulfillment of the project. By way of non-limiting illustration,

an individual project may be associated with a client, and the units of work under the individual project heading may be work directly contributing to the fulfillment of a business relationship with the client.

The values of the project parameters may, by way of non-limiting example, include one or more of: one or more units of work associated with individual ones of the projects (which may include values of other work unit parameters defined by one or more work unit records), status information, user role information, one or more user comment parameters (e.g., a creator, a recipient, one or more followers, one or more other interested parties, content, one or more times, upvotes, other hard-coded responses, etc.), a project name, a project description, one or more project dates (e.g., a start date, a due date, a completion date, and/or other project dates), one or more project collaborators (e.g., an owner, one or more other project collaborators, collaborator access information, and/or other project collaborators and/or collaborator information), one or more attachments, notification settings, privacy, an associated URL, one or more interaction parameters (e.g., sources of the interactions, context of the interactions, content of the interactions, time for the interactions, and/or other interaction parameters), updates, ordering of units of work within the given project, state of a workspace for a given task within the given project, and/or other information.

In some implementations, work unit records may define portfolios. The work unit records defining portfolios may be referred to as portfolio records. The work unit information in portfolio records may be referred to as portfolio information and/or other information. Portfolio information may include values of work unit parameters for portfolios managed within the collaboration environment. The work unit parameters for portfolios may be referred to as “portfolio parameters.” The work unit parameters comprising portfolio parameters may be specific to portfolio records and may include one or more parameters not shared with tasks, projects, objectives, and/or other records. The portfolio parameters may characterize one or more portfolios created, owned, assigned, and/or managed within the collaboration environment and/or via the collaboration environment and/or the metadata associated with the one or more portfolios. Individual ones of the portfolios may be associated with individual ones of the records. A given portfolio may have one or more owners and/or one or more collaborators working on the given portfolio. The given portfolio may be associated with one or more other units of work assigned to one or more users under the given portfolio heading. In some implementations, a given portfolio may not have any owners and/or collaborators at the portfolio level other than those specified with respect to the projects subordinate to the portfolio. In some implementations, portfolios may be associated with one or more units of work that may directly facilitate progress toward fulfillment of the portfolios. Accordingly, completion of units of work may directly contribute to progress toward fulfillment of the portfolio. By way of non-limiting illustration, an individual portfolio may be associated with a client, and the units of work under the individual portfolio heading may be work directly contributing to the fulfillment of a business relationship with the client.

The values of the portfolio parameters may, by way of non-limiting example, include and/or specify one or more of: information about one or more projects, tasks, and/or objectives within a given portfolio, information about one or more other portfolios within a given portfolio (which may include values of portfolio parameters defined by one or

more other portfolio records), a portfolio name, a portfolio description, a status and/or progress (e.g., an update, a hardcoded status update, a measured status, quantity of units of work remaining in a given project, completed units of work in a given project, and/or other status parameter), one or more attachments, notification settings, privacy settings, an associated URL, one or more interaction parameters (e.g., sources of the interactions, context of the interactions, content of the interactions, time for the interactions, and/or other interaction parameters), updates, ordering of projects within the given portfolio, hierarchical information, permissions information, state of a workspace for a given project within the given portfolio, metadata associated with the portfolios, custom fields (e.g., values created by users), and/or other information.

In some implementations, work unit records may define objectives. The work unit records defining objectives may be referred to as objective records. The work unit information in objective records may be referred to as objective information. The objective information may include values of one or more work unit parameters that define the objectives. The values of the work unit parameters may be organized in objective records corresponding to objectives (sometimes referred to as “business objectives”) managed, created, and/or owned within the collaboration environment. The work unit parameters corresponding to objectives may be referred to as “objective parameters.” The work unit parameters comprising objective parameters may be specific to objectives and may include one or more parameters not shared with tasks and/or projects. A given objective may have one or more collaborators, and/or team members working on the given objective. Objectives may be associated with one or more units of work one or more users should accomplish and/or plan on accomplishing. Objectives may be created by a given user for the given user and/or created by the given user and assigned to be owned to one or more other users. Individual objectives may include one or more of an individual goal, an individual sub-goal, and/or other business objectives assigned to be owned by a user and/or associated with one or more users.

The objectives may be associated with a set of units of work that may indirectly facilitate progress toward fulfillment of the objectives. The set of units of work may not directly contribute to the progress. By way of non-limiting illustration, a connection between the set of units of work and a corresponding objective may be indirect in that completion of at least one of the units of work may have no direct impact on progress toward fulfillment of the objective. The concept of “no direct impact” may mean that completion of the at least one unit of work may not cause progress toward fulfillment of the business objective without independent action outside of the at least one unit of work and/or project. Instead, the fulfillment of the at least one unit of work may make such independent action more likely (e.g., through coercion, assistance, education, incentivization, reminder, etc.). However, in some implementations, objectives may be associated with one or more units of work that may directly facilitate progress toward fulfillment of the objectives. Accordingly, completion of the set of units of work may directly contribute to the progress toward fulfillment. Objectives may be associated with an objectives and key result (OKR) goal-setting framework. Objectives may be specified on one or more of a team basis, organization basis, and/or other specifications. In some implementations, objectives may be characterized as user objectives. User objectives may be specified on an individual user basis. A user objective may be associated with a set of units of work

assigned to a user that may indirectly (and/or directly) facilitate progress toward fulfillment of the user objective.

In some implementations, objectives may be described based on one or more of a business objective name, a business objective description, one or more business objective dates (e.g., a start date, a due date, and/or other dates), one or more members associated with a business objective (e.g., an owner, one or more other project/task members, member access information, and/or other business objective members and/or member information), progress information (e.g., an update, a hardcoded status update, a measured status, a progress indicator, quantity value remaining for a given business objective, completed work units in a given project, and/or other progress information), one or more interaction parameters, notification settings, privacy, an associated URL, one or more custom fields (e.g., priority, cost, stage, and/or other custom fields), and/or other information.

The values of the objective owner parameter describing objectives owned by the individual users may be determined based on one or more interactions by one or more users with a collaboration environment. In some implementations, one or more users may create and/or assign ownership of one or more objectives to themselves and/or another user. In some implementations, a user may be assigned to own an objective and the user may effectuate a reassignment of ownership of the objective from the user or one or more other users.

Progress information for the individual objectives may convey progress toward fulfillment of the individual objectives. In some implementations, the progress toward fulfillment of the business objectives may be specified as one or more of a quantitative value, a qualitative value, and/or other information. In some implementations, the quantitative value may be a percentage of completion, an integer value, a dollar amount, and/or other values. In some implementations, progress toward fulfillment of the individual objectives may be determined independently from incremental completion of the units of work associated with the individual objectives. The completion of the units of work associated with a given objective may not directly progress the given business objective toward fulfillment, but completing the units of work may make accomplishing the business objective more likely (e.g., through coercion, assistance, education, incentivization, reminder, etc.). However, in some implementations, progress toward fulfillment of the individual business objectives may be directly determined based on incremental completion of the units of work in the individual sets of units of work associated with the individual business objectives.

Individual sets of work unit records may be defined by a record hierarchy. A record hierarchy may convey individual positions of work unit records (and their corresponding units of work) in the record hierarchy. By way of non-limiting illustration, a position may specify one or more of a work unit record being superior to another work unit record, a work unit record being subordinate to another work unit record, and/or other information. As a result, individual work unit records in the individual sets of work unit records may be subordinate to other individual work unit records in the individual sets of work unit records. For example, a work unit record may define a unit of work comprising a task, and a subordinate work unit record may define a unit of work comprising a sub-task to the task. A record hierarchy may define a relationship between work unit records. A work unit record may have some restrictions placed on it by virtue of having a subordinate work unit record. By way of non-limiting illustration, a work unit record may be restricted

from access (or restricted from marking complete) by one or more users unless and/or until a subordinate work unit record is completed and/or started.

Individual work unit records may include hierarchical information defining one or more record hierarchies of the individual work unit records. The hierarchical information of a work unit record may include one or more of information identifying other work unit records associated in a record hierarchy the work unit record belongs to, a specification of the position of the work unit record in the hierarchy, restrictions and/or other relationships placed on the work unit record by virtue of its position, and/or other information. In some implementations, a hierarchical arrangement may dictate access to different records within the hierarchy based on position and/or other information.

By way of non-limiting illustration, a work unit record may be a portfolio record, and a set of subordinate work unit records may include project records and/or other portfolio records. By way of non-limiting illustration, a work unit record may be a portfolio record, and a set of subordinate work unit records may include project records and task records. By way of non-limiting illustration, a work unit record may be a project record, and a set of subordinate work unit records may include task records.

User role information may specify one or more roles of individual users. A role may represent a position of an individual user. A role may represent one or more of how a user works, how a user is expected to work, how a user intends to work, and/or other considerations. The position may be specified based on a description of one or more of a job title, level, and/or other descriptions of position. A role may be specified objectively with respect to a business organization as a whole and/or other specifications. By way of non-limiting illustration, a role may include one or more of the following: chief executive officer (or other officer), owner, manager, supervisor, accountant, associate, employee, intern, entry level, midlevel, senior, administrator, director, foreman, engineer, product developer, human resource officer, artist, art director, and/or other descriptions.

In some implementations, user role information may specify roles of the users within work unit records. A role may convey expected contribution and/or extent of user interaction of a user in completing and/or supporting a unit of work. By way of non-limiting illustration, a role within a work unit record may include one or more of owner, manager, creator, reviewer, approver, viewer, final decision maker, assignee, assignor, helper, liaison, and/or other descriptions. The individual roles of individual users within some units of work (e.g., a task) may be specified separately from the individual roles of the individual users within other units of work (e.g., a project associated with the task). The individual roles of individual users within the work unit records may be specified separately from the individual roles of the individual users within a business organization as a whole. For example, a business owner may have an "owner" role with respect to a business organization as a whole, while having a "reviewer" role within a project.

In some implementations, individual roles may dictate the type and/or amount of access the users of the individual roles with individual work unit records and/or within the collaboration environment as a whole. By way of non-limiting illustration, users of an "owner" role of a project record may be able to create subordinate work unit records that support fulfillment of the project record, and/or assign other users as assignees to the subordinate work unit records. A user who is an assignee of a subordinate work unit record may be prevented from creating subordinate work unit records that

support fulfillment of the project record, but may be allowed to create sub-subordinate work unit records that support fulfillment of subordinate work unit records they are assigned to. By way of non-limiting illustration, a user of a "viewer" role in a hierarchy of work unit records may only be granted access to view the work unit records in the hierarchy. A user with a "commenter" role within a work unit record may have the ability to add comments to a work unit record, but may not perform other actions such as marking the work unit record as complete, assigning the work unit record to a user, and/or other actions.

In some implementations, environment state information may be continually monitored and/or updated as users interact with the collaboration environment over time. The environment state component **108** may store and/or archive the environment state information continuously, periodically, and/or based on user request to archive. The stored and/or archived environment state information may be referred to as historical environment state information and/or other information.

In some implementations, managing the environment state information by the environment state component **108** may include maintaining queues of work unit records for units of work assigned to the users of a collaboration environment. The queues may be presented to the users in a user interface of the collaboration environment to facilitate access to the work unit records via work unit pages. Individual queues may represent the work unit records assigned to individual users organized in an order based on the individual end dates and/or other dates (e.g., start dates) and/or other ordering. Individual queues may be presented in a user interface based on one or more spatial arrangements. The particular spatial arrangement used by a user may be stored as part of a user record for the user. The spatial arrangements may include one or more of a list view, a calendar view, a board view, and/or other views. The list view may include a vertical arrangement of graphic icons representing work unit records. The calendar view may be a calendar view by week, by more than one week (e.g., 1st through 15th), by month, by more than one month (e.g., May through July), and/or other calendar views. Units of work may be represented in a calendar view by user interface elements (e.g., icons, calendar entries, etc.). The board view may include a grid of cells comprising columns and rows, where visual content items (e.g., icons) representing work unit records may be positioned in the cells. Columns may be associated with sections (e.g., different hierarchies, groupings, etc.), and the cells within the column may be populated with visual content items representing work unit records that are associated with the sections.

In some implementations, environment state component **108** may be configured to generate and/or manage page information defining work unit pages (sometimes referred to simply as "pages") corresponding to user records and/or work unit records. Individual work unit pages may provide access to individual records. Managing information may include one or more of determining, obtaining, receiving, requesting, checking, storing, modifying, and/or other operations. Managing information defining individual work unit pages may include providing information to the user interface component **110** to effectuate presentation of the work unit pages, and/or other information. In some implementations, individual work unit pages may include individual sets of interface elements displaying the values of one or more of the work unit parameters of the individual work unit records.

15

Users may access work unit records via the work unit pages by viewing, adding, changing, deleting, and/or otherwise interacting with values of work unit parameters stored by the work unit records and/or other records. In some implementations, work unit pages providing access to tasks may be referred to as task pages; work unit pages providing access to projects may be referred to as project pages; and work unit pages providing access to objectives may be referred to as objective pages. In some implementations, user pages may include views of the collaboration environment that provide access to user records.

The user interface component **110** may be configured to effectuate communication of interface information to computing platform(s) **104** associated with the users so that the computing platform(s) **104** present instances of a user interface of the collaboration environment. A user interface of the collaboration environment may provide one or more views of the collaboration environment and/or provide other features and/or functionality. The one or more views may include one or more pages of the collaboration environment. The interface information may include page information defining one or more pages of the collaboration environment. In some implementations, a view of the collaboration environment may textually and/or graphically display information (e.g., values of parameters) from one or more of a user record, a task record, a project record, a portfolio record, an objective record, and/or other records. By way of non-limiting illustration, a view may display one or more of a work unit page, a project page, a portfolio page, an objective page, a queue of units of work, and/or other information.

The computing platform(s) **104** may present the work unit pages through which the users access work unit records. Users may provide user input into the work unit pages. The user input may include one or more of selections user interface elements, text additions, drag and drop uploads of digital content items, and/or other user input. Selections may be accomplished through touch-screen input, hovering a cursor and clicking a mouse, entering or typing commands into a command prompt, and/or other types of selections.

The user interface component **110** may be configured to obtain user input information conveying user input into instances of the user interface of the collaboration environment. In some implementations, the user input may include selection of a user interface elements, navigation between pages of a user interface, and/or other input. In some implementations, the type and/or amount of input that is able to be provide into a user interface by a user may be dictated by the access characteristics granted to the user.

The asset group component **112** may be configured to manage asset groups of a computer environment. The individual asset groups may be associated with one or more assets of a computer environment. The individual asset groups may define access characteristics of associated ones of the assets. By way of non-limiting illustration, an asset group may provide a mechanism through which individual users and/or user groups to access one or more assets. That is, an asset group may gate user access to an asset by defining sets of one or more access characteristics to the asset that are able to be realized via the asset group. An asset group may be defined through associations assets, which set out the relationships between the asset groups and the assets. The associations may set out the sets of one or more access characteristics available for an asset through an associated asset group. This may be envisioned as a given asset group comprising a node (e.g., an asset group node), an asset comprising another node (e.g., an asset node), and an edge

16

between the nodes defining a set of one or more access characteristics that are realizable through the asset group (e.g., the asset group-asset relationship). Relationships can define membership for sharing & access control, license assignment from a Billing Account, and more. Asset groups allow the system **100** to represent sets of access characteristics consistently.

By way of non-limiting illustration, the asset groups of a computer environment may include one or more of a first asset group, a second asset group, and/or other asset groups. The first asset group may be associated with a first asset and may define a first set of access characteristics of the first asset of the computer environment. The second asset group may be associated with the first asset and may define a second set of access characteristics of the first asset. By way of non-limiting illustration, in the context of a collaboration environment, the first asset may comprise a first work unit record. The first set of access characteristics may define that “owner,” “viewer,” “commenter,” and “editor” roles are able to be realized for the first work unit record through the first asset group. The second set of access characteristics may define that “manager,” “viewer,” and “editor” roles are able to be realized for the first asset through the second asset group.

The user group component **114** may be configured to manage user groups. The individual user groups may be associated with one or more asset groups. The individual user groups may define the extent of the access characteristics defined by the individual asset groups that are granted to users through the associated individual user groups. The user groups may provide a mechanism to interpret the set of related users and the extent of the access characteristics that are able to be realized through the asset groups are actually granted to the users of the user groups.

In some implementations, the “extent” of the access characteristics are granted to users may be expressed as a whole (e.g., all of) or a part of (e.g., a subset of) the access characteristics of provided through the asset groups.

By way of non-limiting illustration, the user groups may include one or more of a first user group, a second user group, and/or other user groups. The first user group may be associated with the first asset group and may define a first extent of the first set of access characteristics of the first asset that are granted to the users through the first user group. The second user group may be associated with the first asset group and may define a second extent of the first set of access characteristics of the first asset that are granted to the users through the second user group. By way of non-limiting illustration, the first extent of the first set of access characteristics enabled though the first user group may include a first subset of (e.g., a part of) the first set of access characteristics; while the second extent of the first set of access characteristics enabled though the second user group may include the first set of access characteristics (e.g., the whole of the first set).

Again following the collaboration environment example, the first extent may specify that the “owner,” and “editor” roles of the first set of access characteristics are actually granted to the users of the first user group accessing the first asset though the first asset group. The second extent may specify that the “owner,” “viewer,” “commenter,” and “editor” roles of the first set of access characteristics are actually granted to the users of the second user group accessing the first asset though the first asset group. In this example, the second user group grants the users all of the access charac-

17

teristics of the first set of access characteristics, while the first user group grants a subset of the first set of access characteristics.

The environment state component **108** may be configured to assign the users to the user groups. The assignment to the user groups may cause one or more of the access characteristics of associated ones of the assets to be granted to the users based on the extent of the access characteristics that are actually granted to the users through the user groups. By way of non-limiting illustration, a first user may be assigned to the first user group and/or other user groups; a second user may be assigned to the second user group and/or other groups; and/or one or more other users may be assigned to one or more user groups.

The environment state component **108** may be configured to enable and/or disable individual access characteristics of individual assets for individual users of the user groups. Enabling and/or disabling individual access characteristics may be based on assigned ones of the user groups and extent that the access characteristics defined by the asset groups are actually granted to the users through the user groups.

By way of non-limiting illustration, the first extent of the first set of access characteristics may be enabled for the first user accessing the first asset of the computer environment based on the first user being assigned to the first user group and accessing the first asset via the first asset group. By way of non-limiting illustration, the second extent of the first set of access characteristics may be enabled for the second user accessing the first asset based on the second user being assigned to the second user group and accessing the first asset via the first asset group. Again following the above collaboration environment example, the first user may be able to be assigned an "owner" or an "editor" role within the first work unit record, but specifically not a "commenter" or "viewer" role. The second user may be able to be assigned any of the "owner," "viewer," "commenter," and/or "editor" role within the first work unit record.

The environment state component **108** may be configured to identify asset access conflicts. An asset access conflict may arise as a result of an individual user being granted different access characteristics for a common asset. Various situation may occur where this problem arises.

For example, a first asset access conflict situation may occur where 1) a user group is associated with multiple different asset groups, and 2) each of the asset groups define different access characteristics for a common asset. Further, the user group may define different extents in which the different access characteristics are granted to users through the user group via each of the asset groups. That is, the access characteristics granted through the user group via one asset group and the other asset group may differ, thus posing a conflict.

For example, a second asset access conflict situation may occur where 1) a user is associated with multiple user groups, 2) each of the user groups are associated with different asset groups, and 3) each of the asset groups define different access characteristics for a common asset. Further, each of the user groups may define different extents in which the different access characteristics are granted to the user through the different user group via the different asset groups. That is, the access characteristics to a common asset granted to a user through a first user group/asset group combination may be different from the access characteristics to the common asset granted to the user through a second user group/asset group combination, thus posing a conflict.

For example, a third asset access conflict situation may occur where 1) a user is associated with multiple user

18

groups, 2) each of the user groups are associated with a common asset group, 3) the asset group defines access characteristics for an asset, and 4) each of the user groups define different extents in which access characteristics to the asset are granted to users of the different user groups via the common asset group. That is, the access characteristics to the asset granted to users through a first user group/common asset group combination may be different from the access characteristics to the asset granted to the user through a second user group/common asset group combination, thus posing a conflict.

In one or more asset access conflict situations, a question remains of what access characteristics should be granted to user(s) when it comes to accessing a common asset? As described herein, one or more conflict resolution policies may be implemented to handle such situations, including those more complex than described explicitly herein.

By way of non-limiting illustration, considering the first asset access conflict situation described above, the first user group may also be associated with the second asset group. The first user group may define a third extent of the second set of access characteristics of the first asset that are granted to the users of the first user group via the second asset group. In such a situation, environment state component **108** may be configured to identify a first asset access conflict for the users of the first user group accessing the first asset. That is, the first asset access conflict may arise due to the first extent of the first set of access characteristics to the first asset granted through the first user group via the first asset group differing from the third extent of the second set of access characteristics to the first asset granted through the first user group via the second asset group.

For example, considering the collaboration environment situation, the third extent may specify that the "viewer" and "editor" roles of the second set of access characteristics are actually granted to the users of the first user group accessing the first asset through the second asset group. Again, the first user group accessing the first asset through the first asset group tells us that an "owner" and an "editor" role may be assigned within the first work unit record, but not a "commenter" or "viewer" role. However, the first user group accessing the first asset through the second asset group tells us that only the "viewer" and "editor" roles may be assigned within the first work unit record. One specifically excludes the "viewer" role and designates "owner" as an available role; the other specifically grants the "viewer" role, does not designate an "owner" role altogether. A conflict, therefore, exists.

By way of non-limiting illustration, considering the second asset access conflict situation described above, a given user may be included in the first user group and a third user group. The third user group may be associated with the second asset group and may define an extent of the second set of access characteristics of the first asset that are granted to the users through the third user group. In such a situation, the given user accessing the first asset may have two (conflicting) routes to the first asset, e.g., the first user group via the first asset group, and the third user group via the second asset group. The environment state component **108** may be configured to identify an asset access conflict for the given user given this situation.

By way of non-limiting illustration, considering the third asset access conflict situation described above, a given user may be included in the first user group and the second user group. In such a situation, the given user accessing the first asset may have two (conflicting) routes to the first asset, e.g., the first user group via the first asset group, and the second

user group via the first asset group. The environment state component **108** may be configured to identify an asset access conflict for the given user in this situation.

The above descriptions of the first, second, and third asset access conflict situations are provided for illustrative purposes only and are not to be considered limiting as those skilled in the art may appreciate other scenarios where a conflict may arise.

The environment state component **108** may be configured to implement one or more conflict resolution policies to determine which of the access characteristics are to be granted to individual users who are the subject of one or more asset access conflicts. The conflict resolution policies may include rules that allow the conflicts to be resolved. By way of non-limiting illustration, the conflict resolution policies may include one or more of a first conflict resolution policy, a second conflict resolution policy, a third conflict resolution policy, and/or other conflict resolution policies.

In some implementations, the first conflict resolution policy may be associated with giving users of a user group a “maximum” amount of access. Through this policy, a user may be granted more access than would have been granted through any one user/asset group combination. This may include granting all of the access that is being granted through the various asset groups.

In some implementations, the second conflict resolution policy may be associated with giving users of a user group access to the access characteristics that “overlap” between the conflicting groups. Through this policy, a user may be granted access characteristics that are commonly shared between the conflicting groups.

In some implementations, the third conflict resolution policy may be associated with giving users of a user group a “minimum” amount of access. Through this policy, a user may be granted the least amount of access that would have been granted through any of the conflicting groups. This may include granting only the access characteristics of a group that provide the least amount of access.

By way of non-limiting illustration, considering the first asset access conflict situation, the environment state component **108** may be configured to implement the first conflict resolution policy to determine which of the access characteristics are to be granted to the first user. The first conflict resolution policy may result in determining that both the first extent of the first set of access characteristics and the third extent of the second set of access characteristics are granted to the users of the first user group accessing the first asset (e.g., the “maximum”). Continuing with the collaboration environment example, the users of the first user group would be granted the ability to assign the “owner,” “editor,” and “viewer” roles within the first work unit record.

By way of non-limiting illustration, considering the first asset access conflict situation, the environment state component **108** may be configured to implement the second conflict resolution policy to determine which of the access characteristics are to be granted to the first user. The second conflict resolution policy may result in determining that the access characteristics that are commonly shared between the first extent of the first set of access characteristics and the third extent of the second set of access characteristics are granted to the users of the first user group accessing the first asset (e.g., the “overlap”). Continuing with the collaboration environment example, the users of the first user group would be granted the ability to assign the “editor” role, as it is the only role that is commonly shared.

By way of non-limiting illustration, considering the first asset access conflict situation, the environment state com-

ponent **108** may be configured to implement the third conflict resolution policy to determine which of the access characteristics are to be granted to the first user. The third conflict resolution policy may result in determining that either the first extent of the first set of access characteristics or the third extent of the second set of access characteristics is granted to the users of the first user group accessing the first asset (e.g., the “minimum”). Continuing with the collaboration environment example, the users of the first user group would be granted the ability to assign either the “owner” and “editor” roles (e.g., first extent of the first set of access characteristics of the first asset group), or the “viewer” and “editor” roles (e.g., third extent of the second set of access characteristics of the second asset group). In this case, if an “owner” role is considered to have relatively more access to assets than a “viewer,” achieving the “minimum” amount of access through the third conflict resolution policy would mean the “viewer” and “editor” roles would be roles available to the users of the first user group accessing the first work unit record.

FIG. 3 illustrates relationships between user groups, an asset group, and an asset, in accordance with one or more implementations. The graphic in FIG. 3 is intended to be an illustrative example of how different user groups may be granted different access characteristics even though they are accessing a common asset through a common asset group. For example, a first asset group “AG1” may be connected to a first asset “A1” through a first edge “E1.” The first edge E1 may define a relationship between the first asset group AG1 and the first asset A1 in the form of a set of access characteristics of the first asset A1 that are able to be realized through the first asset group AG1. A first user group “UG1” may be connected to the first asset group AG1 through a second edge “E2.” The second edge E2 may define a relationship between the first user group UG1 and the first asset group AG1 as it relates to accessing the first asset A1. By way of non-limiting illustration, the second edge E2 may define an extent of the set of access characteristics that are actually granted to the users of the first user group UG1 accessing the first asset A1 through the first asset group AG1. A second user group “UG2” may be connected to the first asset group AG1 through a third edge “E3.” The third edge E3 may define a relationship between the second user group UG2 and the first asset group AG1 as it relates to accessing the first asset A1. By way of non-limiting illustration, the third edge E3 may define a different extent of the set of access characteristics that are actually granted to the users of the second user group UG2 accessing the first asset A1 through the first asset group AG1.

In FIG. 3, an asset access conflict may arise for a user that is included in both the first user group UG1 and the second user group UG2 and attempting to access first asset A1. For example, the path from UG1-AG1-A1 may result in the realization of different access characteristics than the path UG2-AG1-A1.

FIG. 4 illustrates relationships between user groups, an asset group, and an asset where one or more conflicts may exist, in accordance with one or more implementations. The graphic in FIG. 4 is intended to be an illustrative example of how different user groups may be granted different access characteristics even though they are accessing a common asset through a common asset group. Here, the setup of the first user group UG1, second user group UG2, first asset group AG1, and the first asset A1 may be the same as FIG. 3. However, in FIG. 4, we introduce a second asset group “AG2” connected to the first asset A1 through a fourth edge “E4.” The fourth edge E4 may define a relationship between

21

the second asset group AG2 and the first asset A1 in the form of another set of access characteristics of the first asset A1 that are able to be realized through the second asset group AG2. In this situation, the first user group UG1 may also be connected to the second asset group AG2 through a fifth edge "E5." The fifth edge E5 may define a relationship between the first user group UG1 and the second asset group AG2 as it relates to accessing the first asset A1. By way of non-limiting illustration, the fifth edge E5 may define an extent of the other set of access characteristics that are actually granted to the users of the first user group UG1 accessing the first asset A1 through the second asset group AG2. Because the first user group UG1 has two different paths to the first asset A1—a first path being E2-AG1-E1 and a second path being E5-AG2-E4—a conflict as to what access characteristics the users of the first user group AG1 should get arises. As described herein, one or more conflict resolution policies may be implemented to resolve the conflict.

In FIG. 4, another conflict may arise if we consider the relationship E2 is not present (e.g., the first user group UG1 is only associated with the second asset group AG2). By way of non-limiting illustration, a user who is included in both the first user group UG1 and the second user group UG2 and is attempting to access the first asset A1 may pose a conflict. For example, the path from UG1-AG2-A1 may result in the realization of different access characteristics than the path UG2-AG1-A1.

FIG. 5 illustrates a user interface 500 of a collaboration environment, in accordance with one or more implementations. The user interface 500 may include a view of a collaboration environment. In particular, the user interface 500 may comprise a work unit page 502 for a unit of work through which users access a work unit record for the unit of work. The user interface 500 may display values of one or more work unit parameters and/or other information. By way of non-limiting illustration, a user interface element 501 may display a title of the unit of work. A user interface element 503 may display a due date of unit of work. A user interface element 505 may display an assignee of unit of work. A user interface element or portion 504 may display a description of the unit of work. A user interface element or portion 506 may display a link to a digital content item (e.g., hyperlink to file "Current_design.jpg") and/or other information. A user interface element or portion 508 may display comments added to the work unit page 502. By way of non-limiting illustration, comment 510 may be a comment added by user "Tom"; comment 512 may include a comment added by user "Sue."

The work unit record being accessed through the work unit page 502 may comprise an asset of a collaboration environment. The following are non-limiting examples of the extent of access characteristics that may be granted to the users Tom and Sue, as expressed with respect to interaction with the work unit page 502. Tom may have a first set of access characteristics granted to him for the work unit record that, for example, are an expression of a "commenter" role. Based on this role and associated access characteristics, Tom may be able to view the work unit page 501 and add comments, such as comment 510. However, Tom may be prevented from changing the assignee via element 505, changing the due date via element 503, changing the description via element 504, and/or removing or adding attachments via 506. Sue, on the other hand, may have an "assignee" role, and may be able to view the work unit page 501, add comments such as comment 510, change the due date via element 503, change the description via element

22

504, and/or remove or add attachments via 506. Sue, however, may be prevented from changing the assignee via element 505.

It is noted that the depiction of a work unit page 501 in FIG. 5 and the descriptions of various access characteristics are for illustrative purpose only and not to be considered limiting. Instead, those skilled in the art may appreciate that a work unit page may be displayed in other ways and access characteristics may be expressed differently than described.

Referring back to FIG. 1, in some implementations, server(s) 102, client computing platform(s) 104, and/or external resource(s) 126 may be operatively linked via one or more electronic communication links. For example, such electronic communication links may be established, at least in part, via a network 116 such as the Internet and/or other networks. It will be appreciated that this is not intended to be limiting, and that the scope of this disclosure includes implementations in which server(s) 102, client computing platform(s) 104, and/or external resource(s) 126 may be operatively linked via some other communication media.

A given client computing platform may include one or more processors configured to execute computer program components. The computer program components may be configured to enable an expert or user associated with the given client computing platform to interface with system 100 and/or external resource(s) 126 and/or provide other functionality attributed herein to client computing platform(s) 104. By way of non-limiting example, the given client computing platform 104 may include one or more of a desktop computer, a laptop computer, a handheld computer, a tablet computing platform, a NetBook, a Smartphone, a gaming console, and/or other computing platforms.

External resource(s) 126 may include sources of information outside of system 100, external entities participating with system 100, and/or other resources. In some implementations, some or all the functionality attributed herein to external resource(s) 126 may be provided by resources included in system 100.

Server(s) 102 may include electronic storage 128, one or more processors 130, and/or other components. Server(s) 102 may include communication lines, or ports to enable the exchange of information with a network 116 and/or other computing platforms. Illustration of server(s) 102 in FIG. 1 is not intended to be limiting. Server(s) 102 may include a plurality of hardware, software, and/or firmware components operating together to provide the functionality attributed herein to server(s) 102. For example, server(s) 102 may be implemented by a cloud of computing platforms operating together as server(s) 102.

Electronic storage 128 may comprise non-transitory storage media that electronically stores information. The electronic storage media of electronic storage 128 may include one or both of system storage that is provided integrally (i.e., substantially non-removable) with server(s) 102 and/or removable storage that is removably connectable to server(s) 102 via, for example, a port (e.g., a USB port, a firewire port, etc.) or a drive (e.g., a disk drive, etc.). Electronic storage 128 may include one or more of optically readable storage media (e.g., optical disks, etc.), magnetically readable storage media (e.g., magnetic tape, magnetic hard drive, floppy drive, etc.), electrical charge-based storage media (e.g., EEPROM, RAM, etc.), solid-state storage media (e.g., flash drive, etc.), and/or other electronically readable storage media. Electronic storage 128 may include one or more virtual storage resources (e.g., cloud storage, a virtual private network, and/or other virtual storage resources). Electronic storage 128 may store software algorithms, informa-

tion determined by processor(s) 130, information received from server(s) 102, information received from client computing platform(s) 104, and/or other information that enables server(s) 102 to function as described herein.

Processor(s) 130 may be configured to provide information processing capabilities in server(s) 102. As such, processor(s) 130 may include one or more of a digital processor, a physical processor, an analog processor, a digital circuit designed to process information, an analog circuit designed to process information, a state machine, and/or other mechanisms for electronically processing information. Although processor(s) 130 is shown in FIG. 1 as a single entity, this is for illustrative purposes only. In some implementations, processor(s) 130 may include a plurality of processing units. These processing units may be physically located within the same device, or processor(s) 130 may represent processing functionality of a plurality of devices operating in coordination. Processor(s) 130 may be configured to execute components 108, 110, 112, 114, and/or other components. Processor(s) 130 may be configured to execute components 108, 110, 112, 114 and/or other components by software; hardware; firmware; some combination of software, hardware, and/or firmware; and/or other mechanisms for configuring processing capabilities on processor(s) 130. As used herein, the term "component" may refer to any component or set of components that perform the functionality attributed to the component. This may include one or more physical processors during execution of processor readable instructions, the processor readable instructions, circuitry, hardware, storage media, or any other components.

It should be appreciated that although components 108, 110, 112, and/or 114 are illustrated in FIG. 1 as being implemented within a single processing unit, in implementations in which processor(s) 130 includes multiple processing units, one or more of components 108, 110, 112, and/or 114 may be implemented remotely from the other components. The description of the functionality provided by the different components 108, 110, 112, and/or 114 described below is for illustrative purposes, and is not intended to be limiting, as any of components 108, 110, 112, and/or 114 may provide more or less functionality than is described. For example, one or more of components 108, 110, 112, and/or 114 may be eliminated, and some or all of its functionality may be provided by other ones of components 108, 110, 112, and/or 114. As another example, processor(s) 130 may be configured to execute one or more additional components that may perform some or all of the functionality attributed below to one of components 108, 110, 112, and/or 114.

FIG. 2 illustrates a method 200 to manage access to assets of a computer environment based on user and asset grouping, in accordance with one or more implementations. The operations of method 200 presented below are intended to be illustrative. In some implementations, method 200 may be accomplished with one or more additional operations not described, and/or without one or more of the operations discussed. Additionally, the order in which the operations of method 200 are illustrated in FIG. 2 and described below is not intended to be limiting.

In some implementations, method 200 may be implemented in one or more processing devices (e.g., a digital processor, an analog processor, a digital circuit designed to process information, an analog circuit designed to process information, a state machine, and/or other mechanisms for electronically processing information). The one or more processing devices may include one or more devices executing some or all of the operations of method 200 in response to instructions stored electronically on an electronic storage

medium. The one or more processing devices may include one or more devices configured through hardware, firmware, and/or software to be specifically designed for execution of one or more of the operations of method 200.

An operation 202 may manage asset groups of a computer environment. The individual asset groups may be associated with one or more assets of a computer environment. The individual asset groups may define access characteristics of associated ones of the assets. By way of non-limiting illustration, the asset groups of a computer environment may include a first asset group and/or other asset groups. The first asset group may be associated with a first asset and may define a first set of access characteristics of the first asset of the computer environment. Operation 202 may be performed by one or more hardware processors configured by machine-readable instructions including a component that is the same as or similar to asset group component 112 (shown in FIG. 1 and described herein), in accordance with one or more implementations.

An operation 204 may manage user groups. The individual user groups may be associated with one or more asset groups. The individual user groups may define the extent of the access characteristics defined by the individual asset groups that are granted to users through the associated individual user groups. By way of non-limiting illustration, the user groups may include one or more of a first user group, a second user group, and/or other user groups. The first user group may be associated with the first asset group and may define a first extent of the first set of access characteristics of the first asset that are granted to the users through the first user group. The second user group may be associated with the first asset group and may define a second extent of the first set of access characteristics of the first asset that are granted to the users through the second user group. Operation 204 may be performed by one or more hardware processors configured by machine-readable instructions including a component that is the same as or similar to user group component 114 (shown in FIG. 1 and described herein), in accordance with one or more implementations.

An operation 206 may assign the users to the user groups. The assignment to the user groups may cause one or more of the access characteristics of associated ones of the assets to be granted to the users based on the extent of the access characteristics that are granted to the users through the user groups. By way of non-limiting illustration, a first user may be assigned to the first user group and/or other user groups; a second user may be assigned to the second user group and/or other groups; and/or one or more other users may be assigned to one or more user groups. Operation 206 may be performed by one or more hardware processors configured by machine-readable instructions including a component that is the same as or similar to environment state component 108 (shown in FIG. 1 and described herein), in accordance with one or more implementations.

An operation 208 may enable and/or disable individual access characteristics of individual assets for individual users of the user groups. Enabling and/or disabling individual access characteristics may be based on assigned ones of the user groups and the extent that the access characteristics defined by the asset groups are granted to the users through the user groups. By way of non-limiting illustration, the first extent of the first set of access characteristics may be enabled for the first user accessing the first asset of the computer environment based on the first user being assigned to the first user group. By way of non-limiting illustration, the second extent of the first set of access characteristics may

25

be enabled for the second user accessing the first asset based on the second user being assigned to the second user group. Operation 208 may be performed by one or more hardware processors configured by machine-readable instructions including a component that is the same as or similar to environment state component 108 (shown in FIG. 1 and described herein), in accordance with one or more implementations.

Although the present technology has been described in detail for the purpose of illustration based on what is currently considered to be the most practical and preferred implementations, it is to be understood that such detail is solely for that purpose and that the technology is not limited to the disclosed implementations, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present technology contemplates that, to the extent possible, one or more features of any implementation can be combined with one or more features of any other implementation.

What is claimed:

1. A system configured to manage access to assets of a computer environment based on user and asset grouping, the system comprising:

one or more physical processors configured by machine-readable instructions to:

manage asset groups, where individual asset groups are associated with one or more assets of a computer environment and define access characteristics of associated ones of the one or more assets, the asset groups including a first asset group and a second asset group, the first asset group being associated with a first asset and defining a first set of access characteristics of the first asset, the second asset group being associated with the first asset and defining a second set of access characteristics of the first asset;

manage user groups, where individual user groups are associated with the individual asset groups, the individual user groups defining extent of the access characteristics defined by the individual asset groups that are granted to users through the individual user groups, the user groups including a first user group and a second user group, the first user group being associated with the first asset group and defining a first extent of the first set of access characteristics of the first asset that are granted to the users through the first user group, the second user group being associated with the first asset group and defining a second extent of the first set of access characteristics of the first asset that are granted to the users through the second user group, the first user group also being associated with the second asset group and defining a third extent of the second set of access characteristics of the first asset that are granted to the users through the first user group via the second asset group;

assign the users to the user groups to cause one or more of the access characteristics of the associated ones of the one or more assets to be granted to the users based on the extent of the access characteristics that are granted to the users through the user groups, such that a first user is assigned to the first user group and a second user is assigned to the second user group; enable and/or disable individual access characteristics of individual assets for individual users based on assigned ones of the user groups and the extent that

26

the access characteristics defined by the asset groups are granted to the users through the user groups, such that:

the first extent of the first set of access characteristics is enabled for the first user accessing the first asset based on the first user being assigned to the first user group;

the second extent of the first set of access characteristics is enabled for the second user accessing the first asset based on the second user being assigned to the second user group; and

the third extent of the second set of access characteristics is enabled for the first user accessing the first asset based on the first user being assigned to the first user group;

identify asset access conflicts, including identify a first asset access conflict for the first user with respect to accessing the first asset; and

implement a conflict resolution policy from a set of available conflict resolution policies to determine which of the access characteristics are to be granted to the first user accessing the first asset, the set of available conflict resolution policies including:

a first conflict resolution policy where both the first extent of the first set of access characteristics and the third extent of the second set of access characteristics are granted to the first user accessing the first asset;

a second conflict resolution policy where the access characteristics that are commonly shared between the first extent of the first set of access characteristics and the third extent of the second set of access characteristics are granted to the first user accessing the first asset; and

a third conflict resolution policy where either the first extent of the first set of access characteristics or the third extent of the second set of access characteristics is granted to the first user accessing the first asset.

2. The system of claim 1, wherein the first extent of the first set of access characteristics enabled though the first user group includes a first subset of the first set of access characteristics; and the second extent of the first set of access characteristics enabled though the second user group includes the first set of access characteristics.

3. The system of claim 1, wherein the one or more physical processors are further configured by the machine-readable instructions to:

identify the asset access conflicts as a result of the users of an individual user group being granted different access characteristics for a common asset, such that the first asset access conflict for the users of the first user group is identified by virtue of the first extent of the first set of access characteristics granted through the first user group via the first asset group differing from the third extent of the second set of access characteristics granted through the first user group via the second asset group.

4. The system of claim 1, wherein the access characteristics granted to the users are expressed as one or more of user roles, permissions, license agreements, or control levels.

5. The system of claim 1, wherein the computer environment includes a collaboration environment, and wherein the one or more physical processors are further configured by the machine-readable instructions to:

27

manage environment state information maintaining the collaboration environment, the collaboration environment being configured to facilitate interaction by users with the collaboration environment, the environment state information including work unit records and user records, the work unit records describing units of work created, managed, and/or assigned within the collaboration environment, the user records describing the users; and

wherein an individual asset includes one or more of the work unit records.

6. The system of claim 5, wherein the access characteristics include owner access, editor access, comment access, and view access.

7. The system of claim 5, wherein sets of the work unit records are associated with one or more record hierarchies.

8. A method to manage access to assets of a computer environment based on user and asset grouping, the method comprising:

managing asset groups, where individual asset groups are associated with one or more assets of a computer environment and define access characteristics of associated ones of the one or more assets, the asset groups including a first asset group and a second asset group, the first asset group being associated with a first asset and defining a first set of access characteristics of the first asset, the second asset group being associated with the first asset and defining a second set of access characteristics of the first asset;

managing user groups, where individual user groups are associated with the individual asset groups, the individual user groups defining extent of the access characteristics defined by the individual asset groups that are granted to users through the individual user groups, the user groups including a first user group and a second user group, the first user group being associated with the first asset group and defining a first extent of the first set of access characteristics of the first asset that are granted to the users through the first user group, the second user group being associated with the first asset group and defining a second extent of the first set of access characteristics of the first asset that are granted to the users through the second user group, the first user group also being associated with the second asset group and defining a third extent of the second set of access characteristics of the first asset that are granted to the users through the first user group via the second asset group;

assigning the users to the user groups to cause one or more of the access characteristics of the associated ones of the one or more assets to be granted to the users based on the extent of the access characteristics that are granted to the users through the user groups, including assigning a first user to the first user group and a second user to the second user group; and

enabling and/or disabling individual access characteristics of individual assets for individual users based on assigned ones of the user groups and the extent that the access characteristics defined by the asset groups are granted to the users through the user groups, including: enabling the first extent of the first set of access characteristics for the first user accessing the first asset based on the first user being assigned to the first user group; and

28

enabling the second extent of the first set of access characteristics for the second user accessing the first asset based on the second user being assigned to the second user group; and

enabling the third extent of the second set of access characteristics for the first user accessing the first asset based on the first user being assigned to the first user group;

identifying asset access conflicts, including identifying a first asset access conflict for the first user with respect to accessing the first asset; and

implementing a conflict resolution policy from a set of available conflict resolution policies to determine which of the access characteristics are to be granted to the first user accessing the first asset, the set of available conflict resolution policies including:

a first conflict resolution policy where both the first extent of the first set of access characteristics and the third extent of the second set of access characteristics are granted to the first user accessing the first asset; a second conflict resolution policy where the access characteristics that are commonly shared between the first extent of the first set of access characteristics and the third extent of the second set of access characteristics are granted to the first user accessing the first asset; and a third conflict resolution policy where either the first extent of the first set of access characteristics or the third extent of the second set of access characteristics is granted to the first user accessing the first asset.

9. The method of claim 8, wherein the first extent of the first set of access characteristics enabled though the first user group includes a first subset of the first set of access characteristics; and the second extent of the first set of access characteristics enabled though the second user group includes the first set of access characteristics.

10. The method of claim 8, further comprising:

identifying the asset access conflicts as a result of the users of an individual user group being granted different access characteristics for a common asset, including identifying the first asset access conflict for the users of the first user group by virtue of the first extent of the first set of access characteristics granted through the first user group via the first asset group differing from the third extent of the second set of access characteristics granted through the first user group via the second asset group.

11. The method of claim 8, wherein the access characteristics granted to the users are expressed as one or more of user roles, permissions, license agreements, or control levels.

12. The method of claim 8, wherein the computer environment includes a collaboration environment, and wherein the method further comprises:

managing environment state information maintaining the collaboration environment, the collaboration environment being configured to facilitate interaction by users with the collaboration environment, the environment state information including work unit records and user records, the work unit records describing units of work created, managed, and/or assigned within the collaboration environment, the user records describing the users; and

wherein an individual asset includes one or more of the work unit records.

29

13. The method of claim **12**, wherein the access characteristics include owner access, editor access, comment-only access, and view-only access.

14. The method of claim **12**, wherein sets of the work unit records are associated with one or more record hierarchies. 5

* * * * *

30